

PROJECT UAS

"GlobalScale Retail Data Warehouse"

Disusun Guna Memenuhi Tugas Mata Kuliah

Data Warehouse

Dosen Pengampu :

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**Jurusan Teknologi Informasi
D4 Sistem Informasi Bisnis
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I. STUDI KASUS DAN DATASET

A. Pemilihan Studi Kasus

Studi kasus yang dipilih dalam proyek ini adalah data penjualan ritel dari dataset publik yang tersedia di platform Kaggle dengan judul "Sample Sales Data". Dataset ini dipilih karena memenuhi seluruh kriteria yang dibutuhkan, yaitu kelengkapan atribut data, kemudahan akses, serta ketersediaan secara legal untuk keperluan akademik dan penelitian.

Link Dataset : <https://www.kaggle.com/datasets/kyanyoga/sample-sales-data>

B. Deskripsi Dataset

Tabel dan atribut yang ditemukan dalam dataset tersebut :

1. Tabel dimCustomer

Atribut	Keterangan
id_dimCustomer	Primary Key tabel dimensi pelanggan
customerName	Nama perusahaan/pelanggan
contactFirstName	Nama depan kontak pelanggan
contactLastName	Nama belakang kontak pelanggan
phone	Nomor telepon pelanggan

2. Tabel dimProducts

Atribut	Keterangan
id_dimProduct	Primary Key tabel dimensi produk
productCode	Kode unik produk
productLine	Kategori/lini produk
msrp	Manufacturer's Suggested Retail Price

3. Tabel dimDate

Atribut	Keterangan
id_dimDate	Primary Key tabel dimensi waktu
date	Tanggal transaksi (format lengkap)
month	Bulan transaksi
quarter	Kuartal transaksi
year	Tahun transaksi

4. Tabel dimLocation

Atribut	Keterangan
id_dimLocation	Primary Key tabel dimensi lokasi
addressLine1	Alamat baris pertama
addressLine2	Alamat baris kedua (opsional)
city	Kota
state	Provinsi/negara bagian
postalCode	Kode pos
country	Negara
territory	Wilayah/region

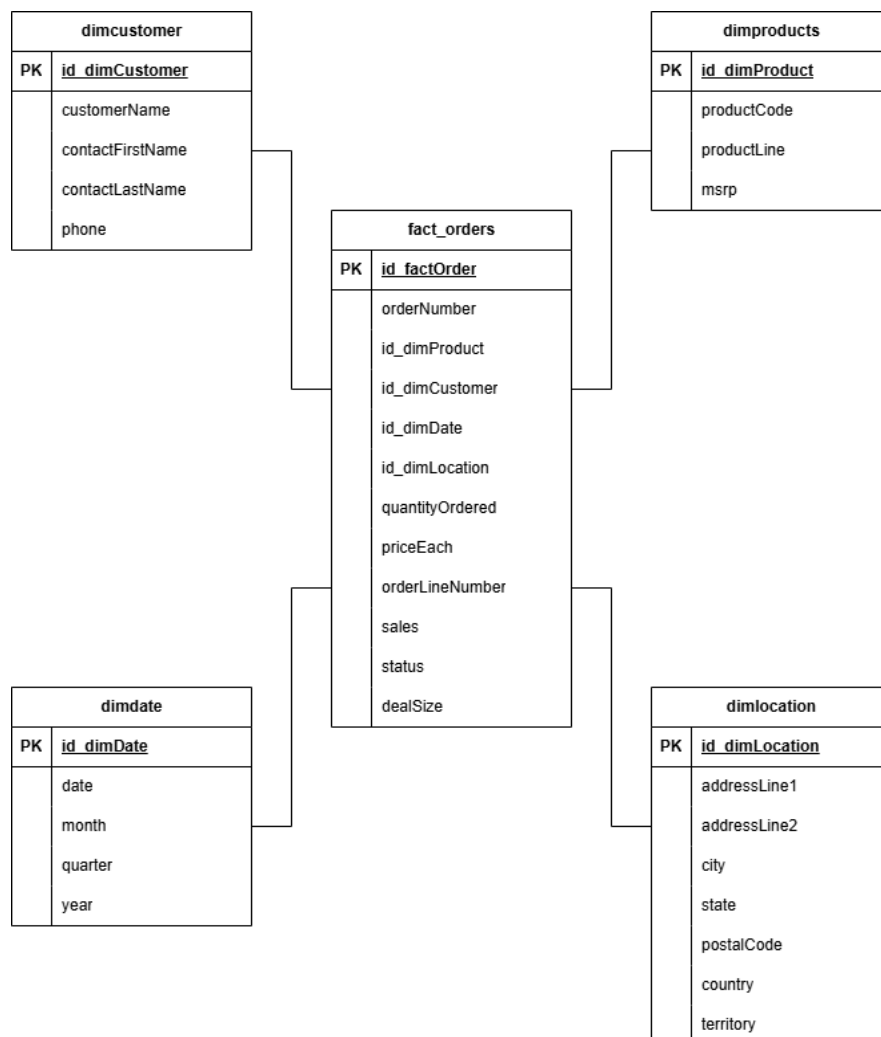
5. Tabel fact_Orders (Tabel Fakta)

Atribut	Keterangan
id_factOrder	Primary Key tabel fakta
orderNumber	Nomor pesanan
id_dimProduct (FK)	Foreign Key ke tabel dimProducts
id_dimCustomer (FK)	Foreign Key ke tabel dimCustomer

id_dimDate (FK)	Foreign Key ke tabel dimDate
id_dimLocation (FK)	Foreign Key ke tabel dimLocation
quantityOrdered	Jumlah item yang dipesan
priceEach	Harga satuan item
orderLineNumber	Nomor baris dalam pesanan
sales	Total nilai penjualan
status	Status pesanan (Shipped, Cancelled, dll.)
dealSize	Ukuran deal (Small, Medium, Large)

II. PERANCANGAN SKEMA BINTANG (STAR SCHEMA)

A. Konsep Star Schema



B. Tabel Fakta : fact_orders

Tabel fakta merupakan inti dari star schema yang menyimpan data numerik terukur (measures) serta foreign key yang menghubungkan ke seluruh tabel dimensi. Tabel fakta dalam proyek ini adalah fact_Orders yang merepresentasikan setiap baris transaksi penjualan.

Atribut ukuran (measures) dalam tabel fakta meliputi:

- quantityOrdered: jumlah produk yang dipesan dalam satu transaksi.
- priceEach: harga satuan produk pada saat transaksi.
- sales: total nilai penjualan (hasil perkalian quantityOrdered dan priceEach).

C. Tabel Dimensi

Tabel dimensi berfungsi sebagai konteks deskriptif bagi data fakta. Berikut adalah empat tabel dimensi yang dirancang :

1. dimCustomer

Menyimpan informasi identitas pelanggan, termasuk nama perusahaan, nama kontak, dan nomor telepon. Dimensi ini memungkinkan analisis penjualan berdasarkan segmen pelanggan.

2. dimProducts

Menyimpan informasi produk, termasuk kode produk, lini produk (kategori), dan harga MSRP. Dimensi ini memungkinkan analisis performa penjualan per produk atau per kategori.

3. dimDate

Menyimpan atribut waktu yang diperlukan untuk analisis temporal, meliputi tanggal, bulan, kuartal, dan tahun. Dimensi waktu merupakan komponen kritis dalam setiap data warehouse karena hampir semua analisis bisnis memiliki dimensi waktu.

4. dimLocation

Menyimpan informasi geografis pengiriman, meliputi alamat, kota, provinsi, kode pos, negara, dan wilayah/territory. Dimensi ini memungkinkan analisis penjualan berdasarkan wilayah geografis.

D. Relasi Antar Tabel

Relasi antar tabel dalam star schema GlobalScale Retail dibangun melalui primary key dan foreign key. Tabel fact_Orders sebagai tabel pusat menyimpan empat foreign key yang masing-masing merujuk ke id_dimCustomer, id_dimProduct, id_dimDate, dan id_dimLocation. Dengan struktur ini, setiap record transaksi dapat ditelusuri secara lengkap berdasarkan identitas pelanggan, produk yang dibeli, waktu transaksi, dan lokasi pengiriman.

III. IMPLEMENTASI PIPELINE ETL

A. Gambaran Umum Pipeline ETL

ETL (Extract, Transform, Load) merupakan proses inti dalam pembangunan data warehouse. Pipeline ETL pada proyek GlobalScale Retail dirancang untuk mengalirkan data dari file CSV sumber ke dalam struktur data warehouse yang telah dirancang. Pipeline ini terdiri dari tiga tahapan utama yang dilakukan secara berurutan.

B. Tahap Extract (Ekstraksi Data)

1. Dataset sales_data_sample.csv dibuka di Excel, terdiri dari 2824 baris data dan 25 kolom yaitu ORDERNUMBER,QUANTITYORDERED, PRICEEACH,ORDERLINENUMBER,SALES,ORDERDATE, STATUS,QTR_ID,MONTH_ID,YEAR_ID,PRODUCTLINE, MSRP,PRODUCTCODE,CUSTOMERNAME,PHONE, ADDRESSLINE1,ADDRESSLINE2,CITY,STATE,POSTALCODE, COUNTRY,TERRITORY,CONTACTLASTNAME,CONTACTFIRST NAME, DEALSIZE.

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	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
	ORDERDATE	STATUS	QTR_ID	MONTH_1	YEAR_ID	PRODUCTLINE	MSRP	PRODUCTCODE	CUSTOMERNAME	PHONE	ADDRESSLINE1	ADDRESSLINE2	CITY	STATE	POSTAL
1	2/24/2003 0:00	Shipped	1	2	2003	Motorcycles	\$5 110_1678		Land of Toys Inc.	2155557818	897 Long Port Avenue		NYC	NY	
2	5/7/2003 0:00	Shipped	2	5	2003	Motorcycles	\$5 110_1678		Reims Collectables	26.47.1555	59 rue de l'Abbaye		Reims		
3	7/17/2003 0:00	Shipped	3	7	2003	Motorcycles	\$5 110_1678		Lion Souveniers	+33 1 46 62 757 127	rue du Colonel Pierre Avia		Paris		
4	8/25/2003 0:00	Shipped	4	8	2003	Motorcycles	\$5 110_1678		Toys4townpeople.com	6265552265	78934 Villado Dr.		Palmdale	CA	91367
5	10/1/2003 0:00	Shipped	4	10	2003	Motorcycles	\$5 110_1678		Corporate Gift Ideas Co.	6505551386	7734 Strong St.		San Francisco	CA	94133
6	10/28/2003 0:00	Shipped	4	10	2003	Motorcycles	\$5 110_1678		Theatrical Stores Inc.	6505556809	9408 Fourth Circle		Burlington	CA	94401
7	11/11/2003 0:00	Shipped	4	11	2003	Motorcycles	\$5 110_1678		Daedalus Designs Imports	20.16.1555	184, chaussee de Tournai		Lille		
8	11/18/2003 0:00	Shipped	4	11	2003	Motorcycles	\$5 110_1678		Heriku Gifts	+47 2267 3215	Drummen 121, PB 744 Sentrum		Bergen	N	5804
9	12/1/2003 0:00	Shipped	4	12	2003	Motorcycles	\$5 110_1678		Mini Wheels Co.	6505555787	5557 North Pendle Street		Perlis	San Francisco	CA
10	1/15/2004 0:00	Shipped	1	1	2004	Motorcycles	\$5 110_1678		Auto Canal Petit	(1) 47.56.6555	25, rue Lauriston	Level 3	Boris		
11	2/20/2004 0:00	Shipped	1	2	2004	Motorcycles	\$5 110_1678		Australian Collectors, Co.	93 9520 4555	636 St Kilda Road		Melbourne	VIC	
12	4/5/2004 0:00	Shipped	2	4	2004	Motorcycles	\$5 110_1678		Viachrome Inc.	6175555500	2678 Kingston Rd.	Suite 101	NYC	NY	
13	5/18/2004 0:00	Shipped	2	5	2004	Motorcycles	\$5 110_1678		Toy Collectables Inc.	2015550350	7476 Moss Rd.		Newark	NJ	
14	6/28/2004 0:00	Shipped	2	6	2004	Motorcycles	\$5 110_1678		Gift Depot Inc.	2035552570	25593 South Bay Ln.		Bridgewater	CT	
15	7/23/2004 0:00	Shipped	3	7	2004	Motorcycles	\$5 110_1678		La Rochelle Gifts	40.67.8555	67, rue des Cinqante Otages		Nantes		
16	8/27/2004 0:00	Shipped	3	8	2004	Motorcycles	\$5 110_1678		Marta's Replicas Co.	6175559555	39523 Spinnaker Dr.		Cambridge MA		
17	9/30/2004 0:00	Shipped	3	9	2004	Motorcycles	\$5 110_1678		Toys of Finland, Co.	90.224 8555	Keskuskatu		Helsinki		
18	10/15/2004 0:00	Shipped	4	10	2004	Motorcycles	\$5 110_1678		Baane Mini Replicas Co.	97.98 9555	Erfing Skakke gate 78		Stavern		
19	11/2/2004 0:00	Shipped	4	11	2004	Motorcycles	\$5 110_1678		Diecast Classics Inc.	2155551555	7586 Pompton St.		Allentown PA		
20	11/15/2004 0:00	Shipped	4	11	2004	Motorcycles	\$5 110_1678		Land of Toys Inc.	2155557818	897 Long Port Avenue		NYC	NY	
21	12/14/2004 0:00	Shipped	4	12	2004	Motorcycles	\$5 110_1678		Sailburg Collectables	6562 9555	Gieslweg 14		Salzburg		
22	12/17/2004 0:00	Shipped	4	12	2004	Motorcycles	\$5 110_1678		Souvenirs And Things Co.	+61 2 9495 8555	Monitor Movie Building, 815 Pac Level 6	Level 6	Chatswood NSW		
23	2/13/2005 0:00	Shipped	1	2	2005	Motorcycles	\$5 110_1678		La Rochelle Gifts	40.67.8555	67, rue des Cinqante Otages		Nantes		

< sales_data_sample >

2. Identifikasi missing values

Dari hasil inspeksi awal ditemukan 4 kolom yang memiliki nilai kosong yaitu ADDRESSLINE2 sebanyak 2521 baris, STATE sebanyak 1486 baris, POSTALCODE sebanyak 76 baris, dan TERRITORY sebanyak 1074 baris.

3. Identifikasi duplikasi

Dari hasil pengecekan duplikasi tidak ditemukan baris data yang duplikat.

C. Inisialisasi Database & Skema DDL (Data Definition Language)

1. Membuat Database

```
CREATE DATABASE IF NOT EXISTS dw_sales;  
USE dw_sales;
```

2. Membuat Tabel Dimensi (Dimension Tables)

-- Tabel Dimensi: dimcustomer

```
CREATE TABLE IF NOT EXISTS dimcustomer (  
    id_dimCustomer INT NOT NULL AUTO_INCREMENT,  
    customerName VARCHAR(50) NOT NULL,  
    contactFirstName VARCHAR(50) NOT NULL,  
    contactLastName VARCHAR(50) NOT NULL,  
    phone VARCHAR(20) NOT NULL,  
    PRIMARY KEY (id_dimCustomer)  
);
```

-- Tabel Dimensi: dimproducts

```
CREATE TABLE IF NOT EXISTS dimproducts (  
    id_dimProduct INT NOT NULL AUTO_INCREMENT,  
    productCode VARCHAR(15) NOT NULL,  
    productLine VARCHAR(50) NOT NULL,  
    msrp DECIMAL(10,2) NOT NULL,  
    PRIMARY KEY (id_dimProduct)  
);
```

-- Tabel Dimensi: dimdate

```
CREATE TABLE IF NOT EXISTS dimdate (  
    id_dimDate INT NOT NULL AUTO_INCREMENT,  
    date       DATE NOT NULL,  
    month      INT NOT NULL,  
    quarter    INT NOT NULL,  
    year       INT NOT NULL,  
    PRIMARY KEY (id_dimDate)  
);
```

-- Tabel Dimensi: dimlocation

```
CREATE TABLE IF NOT EXISTS dimlocation (  
    id_dimLocation INT NOT NULL AUTO_INCREMENT,  
    addressLine1   VARCHAR(100) NOT NULL,  
    addressLine2   VARCHAR(100),  
    city           VARCHAR(50) NOT NULL,  
    state          VARCHAR(50),  
    postalCode     VARCHAR(15),  
    country        VARCHAR(50) NOT NULL,  
    territory      VARCHAR(50),  
    PRIMARY KEY (id_dimLocation)  
);
```

3. Membuat Tabel Fakta dengan Relasi Foreign Key (Fact Table)

-- Tabel Fakta: fact_orders

```
CREATE TABLE IF NOT EXISTS fact_orders (  
    id_factOrder INT NOT NULL AUTO_INCREMENT,  
    orderNumber  INT NOT NULL,  
    id_dimProduct INT NOT NULL,  
    id_dimCustomer INT NOT NULL,  
    id_dimDate   INT NOT NULL,  
    id_dimLocation INT NOT NULL,  
    quantityOrdered INT NOT NULL,  
    priceEach    DECIMAL(10,2) NOT NULL,
```

```

orderLineNumber INT NOT NULL,
sales DECIMAL(10,2) NOT NULL,
status VARCHAR(15) NOT NULL,
dealSize VARCHAR(10) NOT NULL,
PRIMARY KEY (id_factOrder),
FOREIGN KEY (id_dimProduct) REFERENCES
dimproducts(id_dimProduct),
FOREIGN KEY (id_dimCustomer) REFERENCES
dimcustomer(id_dimCustomer),
FOREIGN KEY (id_dimDate) REFERENCES
dimdate(id_dimDate),
FOREIGN KEY (id_dimLocation) REFERENCES
dimlocation(id_dimLocation)
);

```

4. Optimalisasi dan Standardisasi Charset Database

Run SQL query/queries on database **dw_sales**:

```

1 ALTER DATABASE dw_sales CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
2 ALTER TABLE dimcustomer CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
3 ALTER TABLE dimproducts CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
4 ALTER TABLE dimdate CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
5 ALTER TABLE dimlocation CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
6 ALTER TABLE fact_orders CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0012 seconds.)

ALTER DATABASE dw_sales CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0078 seconds.)

ALTER TABLE dimcustomer CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0119 seconds.)

ALTER TABLE dimproducts CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0038 seconds.)

ALTER TABLE dimdate CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0092 seconds.)

ALTER TABLE dimlocation CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

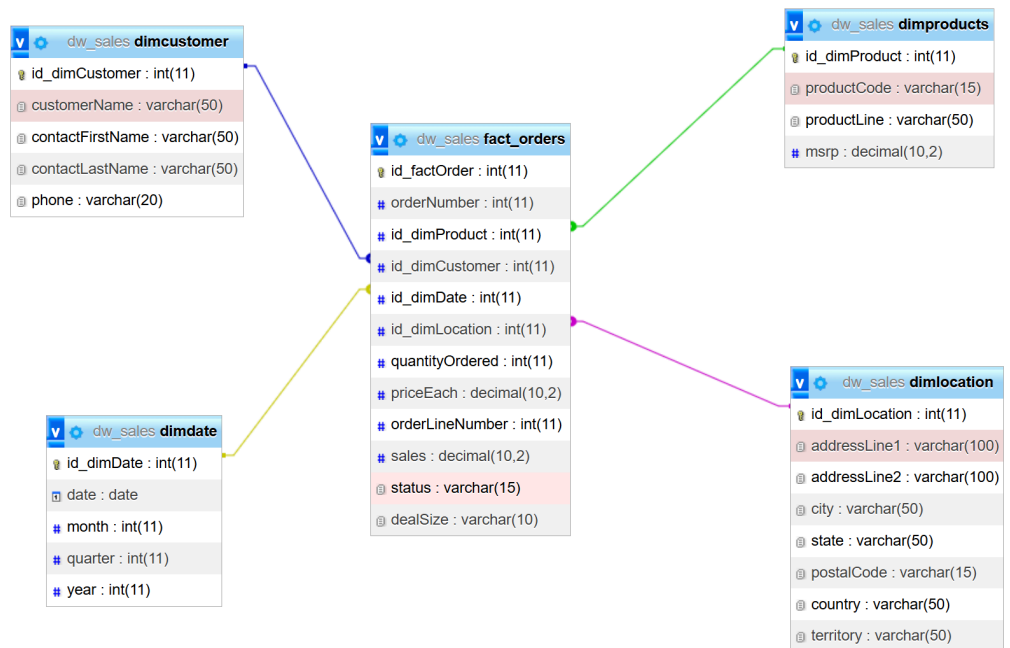
[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0037 seconds.)

ALTER TABLE fact_orders CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

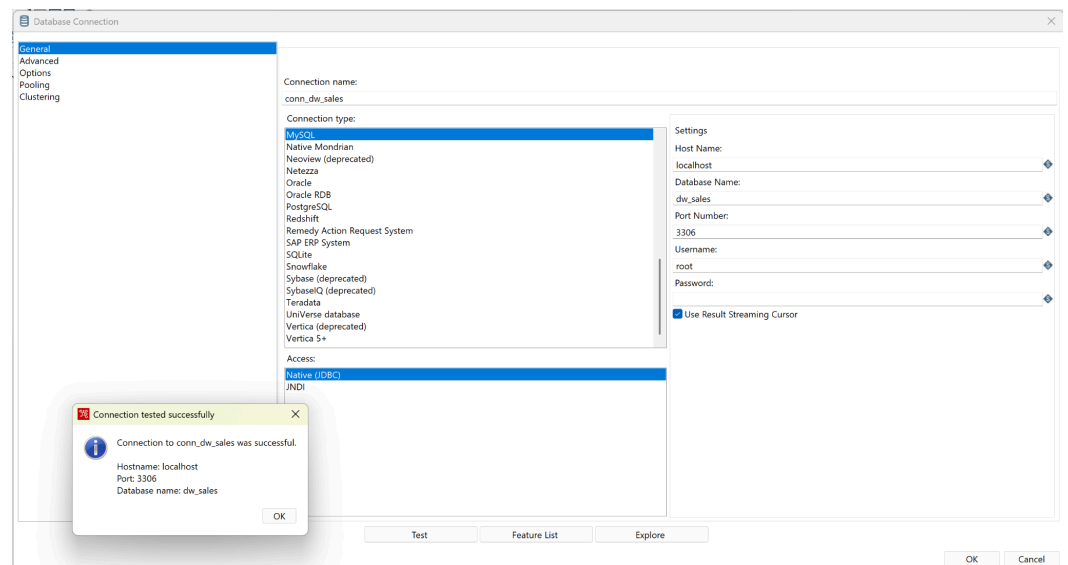
[Edit inline] [Edit] [Create PHP code]

5. Design Database



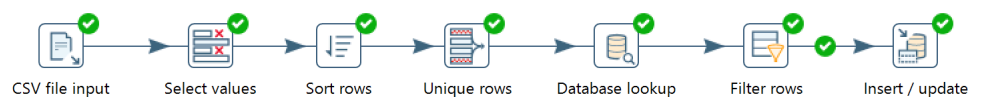
D. Tahap Transform (Transformasi Data)

1. Create connection database



2. Transformasi dim Products

1. Membuat object



2. Konfigurasi CSV file input

Step name: CSV file input

Filename: D:\College\Semester 4\Data Warehouse\dataset\sales_data_sample.csv

Delimiter: ,

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result? ☐

The row number field name (optional): QTR_ID

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding: mixed

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	ORDERNUMBER	Integer	#	15	0	\$	.	.	none
2	QUANTITYORDERED	Integer	#	15	0	\$	.	.	none
3	PRICEEACH	BigNumber	###	5	2	\$	.	.	none
4	ORDERLINENUMBER	Integer	#	15	0	\$	.	.	none
5	SALES	BigNumber	###	7	2	\$	.	.	none
6	ORDERDATE	String		15		\$	.	.	none
7	STATUS	String		10		\$	.	.	none
8	QTR_ID	Integer	#	15	0	\$	.	.	none
9	MONTH_ID	Integer	#	15	0	\$	.	.	none
10	YEAR_ID	Integer	#	15	0	\$	.	.	none
11	PRODUCTLINE	String		12		\$	.	.	none
12	MSRP	Integer	#	15	0	\$	.	.	none
13	PRODUCTCODE	String		8		\$	.	.	none
14	CUSTOMERNAME	String		30		\$	.	.	none
15	PHONE	String		16		\$	.	.	none
16	ADDRESSLINE1	String		40		\$	.	.	none
17	ADDRESSLINE2	String		9		\$	.	.	none
18	CITY	String		14		\$	.	.	none
19	STATE	String		10		\$	.	.	none
20	POSTALCODE	String		8		\$	.	.	none
21	COUNTRY	String		9		\$	.	.	none
22	TERRITORY	String		5		\$	.	.	none
23	CONTACTLASTNAME	String		10		\$	.	.	none
24	CONTACTFIRSTNAME	String		9		\$	.	.	none
25	DEALSIZE	String		6		\$	.	.	none

3. Select values

- Select & alter

Step name: Select values

Select & Alter Remove Meta-data

Fields:

#	Fieldname	Rename to	Length	Precision
1	ORDERNUMBER			
2	QUANTITYORDERED			
3	PRICEEACH			
4	ORDERLINENUMBER			
5	SALES			
6	ORDERDATE			
7	STATUS			
8	QTR_ID			
9	MONTH_ID			
10	YEAR_ID			
11	PRODUCTLINE	productLine		
12	MSRP	msrp		
13	PRODUCTCODE	productCode		
14	CUSTOMERNAME			
15	PHONE			
16	ADDRESSLINE1			
17	ADDRESSLINE2			
18	CITY			
19	STATE			
20	POSTALCODE			
21	COUNTRY			
22	TERRITORY			
23	CONTACTLASTNAME			
24	CONTACTFIRSTNAME			
25	DEALSIZE			

Include unspecified fields, ordered by name ☐

Get fields to select Edit Mapping

- Remove

The 'Select values' dialog box is shown with the 'Remove' tab selected. The 'Step name' is 'Select values'. The 'Fields to remove' table lists 22 fields. A 'Get fields to remove' button is on the right. At the bottom are 'Help', 'OK', and 'Cancel' buttons.

#	Fieldname
1	ORDERNUMBER
2	QUANTITYORDERED
3	PRICEEACH
4	ORDERLINENUMBER
5	SALES
6	ORDERDATE
7	STATUS
8	QTR_ID
9	MONTH_ID
10	YEAR_ID
11	CUSTOMERNAME
12	PHONE
13	ADDRESSLINE1
14	ADDRESSLINE2
15	CITY
16	STATE
17	POSTALCODE
18	COUNTRY
19	TERRITORY
20	CONTACTLASTNAME
21	CONTACTFIRSTNAME
22	DEALSIZE

4. Sort rows

The 'Sort rows' dialog box is shown. The 'Step name' is 'Sort rows'. The 'Sort directory' is '%%java.io.tmpdir%%'. The 'TMP-file prefix' is 'out'. The 'Sort size (rows in memory)' is '1000000'. The 'Free memory threshold (in %)' is empty. The 'Compress TMP Files?' checkbox is checked. The 'Only pass unique rows? (verifies keys only)' checkbox is unchecked. The 'Fields' table has 7 columns and 1 row. At the bottom are 'Help', 'OK', 'Cancel', and 'Get Fields' buttons.

#	Fieldname	Ascending	Case sensitive compare?	Sort based on current locale?	Collator Strength	Presorted?
1	productCode	Y	N	N	0	N

5. Unique rows

Unique rows

Step name Unique rows

Settings

Add counter to output? ☐ Counter field

Redirect duplicate row ☐ Error description

Fields to compare on (no entries means: compare complete row)

#	Fieldname	Ignore case
1	productCode	N

6. Database lookup

Database lookup

Step name Database lookup

Connection conn_dw_sales

Lookup schema dw_sales

Lookup table dimproducts

Enable cache? ☐

Cache size in rows (0=cache) 0

Load all data from table ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	productCode	=	productCode	

Values to return from the lookup table :

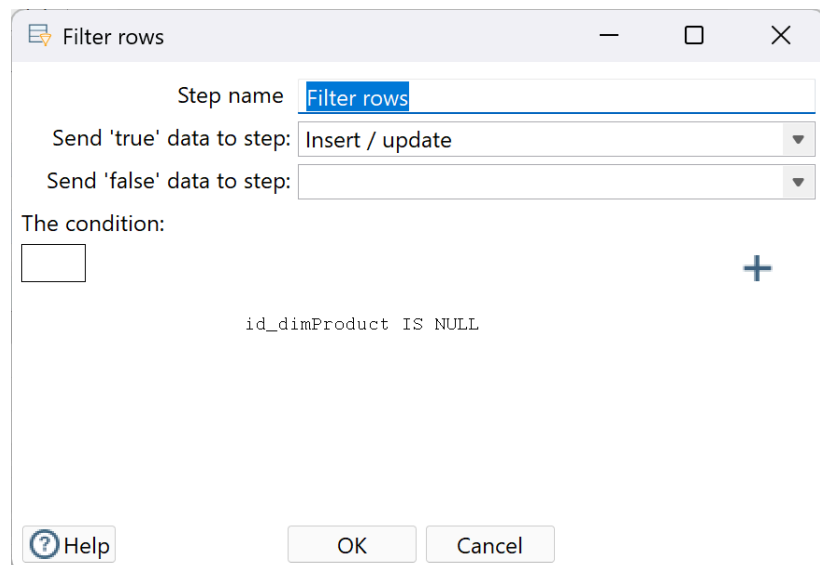
#	Field	New name	Default	Type
1	id_dimProduct			Integer

Do not pass the row if the lookup fails ☐

Fail on multiple results? ☐

Order by

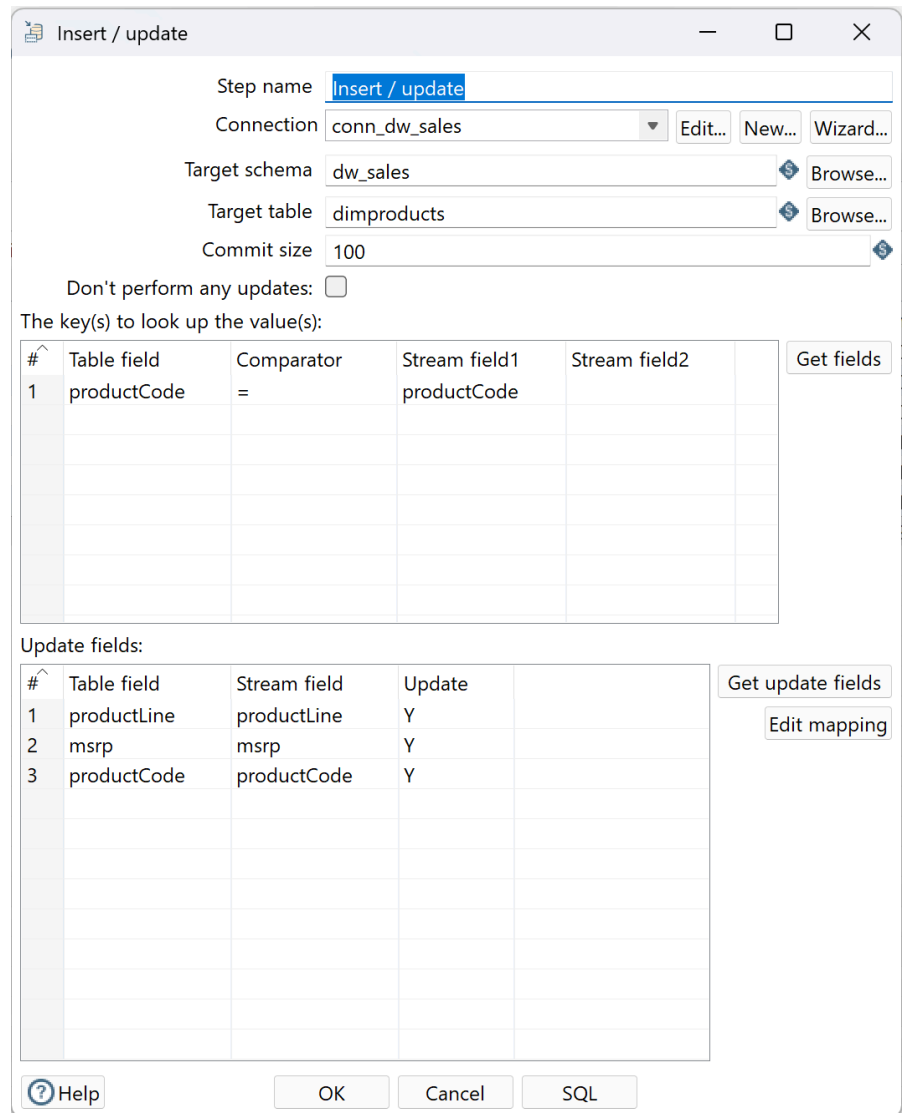
7. Filter rows



Filter rows dialog box configuration:

- Step name:
- Send 'true' data to step:
- Send 'false' data to step:
- The condition:
- Buttons: Help, OK, Cancel

8. Insert / Update



Insert / update dialog box configuration:

- Step name:
- Connection: - Target schema: - Target table: - Commit size:
- Don't perform any updates: ☐
- The key(s) to look up the value(s):

#	Table field	Comparator	Stream field1	Stream field2
1	productCode	=	productCode	
- Update fields:

#	Table field	Stream field	Update
1	productLine	productLine	Y
2	msrp	msrp	Y
3	productCode	productCode	Y
- Buttons: Help, OK, Cancel, SQL

9. Hasil run

The image displays the execution results of a data transformation job in Apache Spark, showing a sequence of steps from CSV file input to insert/update.

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

2026-06-03 20:41:40.442 - Spoon - Save file as...
2026-06-03 20:41:40.477 - Spoon - Transformation opened.
2026-06-03 20:41:40.477 - Spoon - Launching transformation [dimproducts_project uas]...
2026-06-03 20:41:40.477 - Spoon - Started the transformation execution.
2026-06-03 20:41:40.797 - dimproducts_project uas - Dispatching started for transformation [dimproducts_project uas]
2026-06-03 20:41:40.846 - CSV file input.0 - Header row skipped in file 'D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv'
2026-06-03 20:41:40.853 - CSV file input.0 - Finished processing (I=2824, O=0, R=0, W=2823, U=0, E=0)
2026-06-03 20:41:40.857 - Select values.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 20:41:40.878 - Sort rows.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 20:41:40.896 - Unique rows.0 - Finished processing (I=0, O=0, R=2823, W=109, U=0, E=0)
2026-06-03 20:41:40.937 - Database lookup.0 - Finished processing (I=0, O=0, R=109, W=109, U=0, E=0)
2026-06-03 20:41:40.956 - Filter rows.0 - Finished processing (I=0, O=0, R=109, W=109, U=0, E=0)
2026-06-03 20:41:41.020 - Insert / update.0 - Finished processing (I=109, O=109, R=109, W=109, U=0, E=0)
2026-06-03 20:41:41.021 - Spoon - The transformation has finished!!

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	2823	2824	0	0	0	0	0	0.0s	282,400	-
2	Select values	0	2823	2823	0	0	0	0	0	0	0.0s	201,643	-
3	Sort rows	0	2823	2823	0	0	0	0	0	0	0.0s	78,417	-
4	Unique rows	0	2823	109	0	0	0	0	0	0	0.1s	52,278	-
5	Database lookup	0	109	109	0	0	0	0	0	0	0.1s	1,160	-
6	Filter rows	0	109	109	0	0	0	0	0	0	0.1s	965	-
7	Insert / update	0	109	109	109	109	0	0	0	0	0.2s	612	-

Showing rows 0 - 24 (109 total, Query took 0.0007 seconds.) [id_dimProduct: 109... - 85...]

```
SELECT * FROM `dimproducts` ORDER BY `dimproducts`.`id_dimProduct` DESC
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Number of rows: 25 Filter rows: Search this table

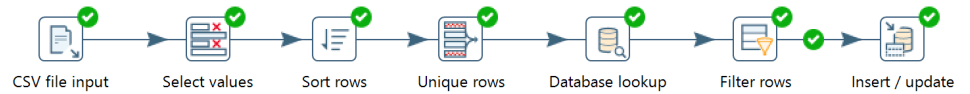
Extra options

	id_dimProduct	productCode	productLine	msrp
☐ Edit ☐ Copy ☐ Delete	109	S72_3212	Ships	54.00
☐ Edit ☐ Copy ☐ Delete	108	S72_1253	Planes	49.00
☐ Edit ☐ Copy ☐ Delete	107	S700_4002	Planes	74.00
☐ Edit ☐ Copy ☐ Delete	106	S700_3962	Ships	99.00
☐ Edit ☐ Copy ☐ Delete	105	S700_3505	Ships	100.00
☐ Edit ☐ Copy ☐ Delete	104	S700_3167	Planes	80.00
☐ Edit ☐ Copy ☐ Delete	103	S700_2834	Planes	118.00
☐ Edit ☐ Copy ☐ Delete	102	S700_2824	Classic Cars	101.00
☐ Edit ☐ Copy ☐ Delete	101	S700_2610	Ships	72.00
☐ Edit ☐ Copy ☐ Delete	100	S700_2466	Planes	99.00
☐ Edit ☐ Copy ☐ Delete	99	S700_2047	Ships	90.00
☐ Edit ☐ Copy ☐ Delete	98	S700_1938	Ships	86.00
☐ Edit ☐ Copy ☐ Delete	97	S700_1691	Planes	91.00
☐ Edit ☐ Copy ☐ Delete	96	S700_1138	Ships	66.00
☐ Edit ☐ Copy ☐ Delete	95	S50_4713	Motorcycles	81.00
☐ Edit ☐ Copy ☐ Delete	94	S50_1514	Trains	58.00
☐ Edit ☐ Copy ☐ Delete	93	S50_1392	Trucks and Buses	115.00
☐ Edit ☐ Copy ☐ Delete	92	S50_1341	Vintage Cars	43.00
☐ Edit ☐ Copy ☐ Delete	91	S32_4485	Motorcycles	102.00
☐ Edit ☐ Copy ☐ Delete	90	S32_4289	Vintage Cars	68.00

Console

3. Transformasi dim Customer

1. Membuat object



2. Konfigurasi CSV file input

Step name: CSV file input

Filename: D:\College\Semester 4\Data Warehouse\dataset\sales_data_sample.csv

Delimiter: ;

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result: ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in field? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	ORDERNUMBER	Integer	#	15	0	\$	-	.	none
2	QUANTITYORDERED	Integer	#	15	0	\$	-	.	none
3	PRICEEACH	BigNumber	#,##	5	2	\$	-	.	none
4	ORDERLINENUMBER	Integer	#	15	0	\$	-	.	none
5	SALES	BigNumber	#,##	7	2	\$	-	.	none
6	ORDERDATE	String		15		\$	-	.	none
7	STATUS	String		10		\$	-	.	none
8	QTR_ID	Integer	#	15	0	\$	-	.	none
9	MONTH_ID	Integer	#	15	0	\$	-	.	none
10	YEAR_ID	Integer	#	15	0	\$	-	.	none
11	PRODUCTLINE	String		12		\$	-	.	none
12	MSRP	Integer	#	15	0	\$	-	.	none
13	PRODUCTCODE	String		8		\$	-	.	none
14	CUSTOMERNAME	String		30		\$	-	.	none
15	PHONE	String		16		\$	-	.	none
16	ADDRESSLINE1	String		40		\$	-	.	none
17	ADDRESSLINE2	String		9		\$	-	.	none
18	CITY	String		14		\$	-	.	none
19	STATE	String		10		\$	-	.	none
20	POSTALCODE	String		8		\$	-	.	none
21	COUNTRY	String		9		\$	-	.	none
22	TERRITORY	String		5		\$	-	.	none
23	CONTACTLASTNAME	String		10		\$	-	.	none
24	CONTACTFIRSTNAME	String		9		\$	-	.	none
25	DEALSIZE	String		6		\$	-	.	none

3. Select values

- Select & alter

Step name: Select values

Select & Alter Remove Meta-data

Fields:

#	Fieldname	Rename to	Length	Precision
1	ORDERNUMBER			
2	QUANTITYORDERED			
3	PRICEEACH			
4	ORDERLINENUMBER			
5	SALES			
6	ORDERDATE			
7	STATUS			
8	QTR_ID			
9	MONTH_ID			
10	YEAR_ID			
11	PRODUCTLINE			
12	MSRP			
13	PRODUCTCODE			
14	CUSTOMERNAME	customerName		
15	PHONE	phone		
16	ADDRESSLINE1			
17	ADDRESSLINE2			
18	CITY			
19	STATE			
20	POSTALCODE			
21	COUNTRY			
22	TERRITORY			
23	CONTACTLASTNAME	contactLastName		
24	CONTACTFIRSTNAME	contactFirstName		
25	DEALSIZE			

Get fields to select

Edit Mapping

Include unspecified fields, ordered by name ☐

OK Cancel

- Remove

The 'Select values' dialog box is shown with the 'Remove' tab selected. The 'Step name' is 'Select values'. The 'Fields to remove' table lists 21 fields. A 'Get fields to remove' button is on the right. At the bottom are 'Help', 'OK', and 'Cancel' buttons.

#	Fieldname
1	ORDERNUMBER
2	QUANTITYORDERED
3	PRICEEACH
4	ORDERLINENUMBER
5	SALES
6	ORDERDATE
7	STATUS
8	QTR_ID
9	MONTH_ID
10	YEAR_ID
11	PRODUCTLINE
12	MSRP
13	PRODUCTCODE
14	ADDRESSLINE1
15	ADDRESSLINE2
16	CITY
17	STATE
18	POSTALCODE
19	COUNTRY
20	TERRITORY
21	DEALSIZE

4. Sort rows

The 'Sort rows' dialog box is shown. The 'Step name' is 'Sort rows'. The 'Sort directory' is '%%java.io.tmpdir%%'. The 'TMP-file prefix' is 'out'. The 'Sort size (rows in memory)' is '1000000'. The 'Free memory threshold (in %)' is empty. The 'Compress TMP Files?' checkbox is checked. The 'Only pass unique rows? (verifies keys only)' checkbox is unchecked. The 'Fields' table lists one field: 'customerName'. At the bottom are 'Help', 'OK', 'Cancel', and 'Get Fields' buttons.

#	Fieldname	Ascending	Case sensitive compare?	Sort based on current locale?	Collator Strength	Presorted?
1	customerName	Y	N	N	0	N

5. Unique rows

Unique rows

Step name Unique rows

Settings

Add counter to output? ☐ Counter field

Redirect duplicate row ☐ Error description

Fields to compare on (no entries means: compare complete row)

#	Fieldname	Ignore case
1	customerName	N

[? Help](#) [OK](#) [Cancel](#) [Get](#)

6. Database lookup

Database lookup

Step name Database lookup

Connection conn_dw_sales [Edit...](#) [New...](#) [Wizard...](#)

Lookup schema dw_sales [Browse...](#)

Lookup table dimcustomer [Browse...](#)

Enable cache? ☐

Cache size in rows (0=cache) 0

Load all data from table ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	customerName	=	customerName	

Values to return from the lookup table :

#	Field	New name	Default	Type
1	id_dimCustomer			Integer

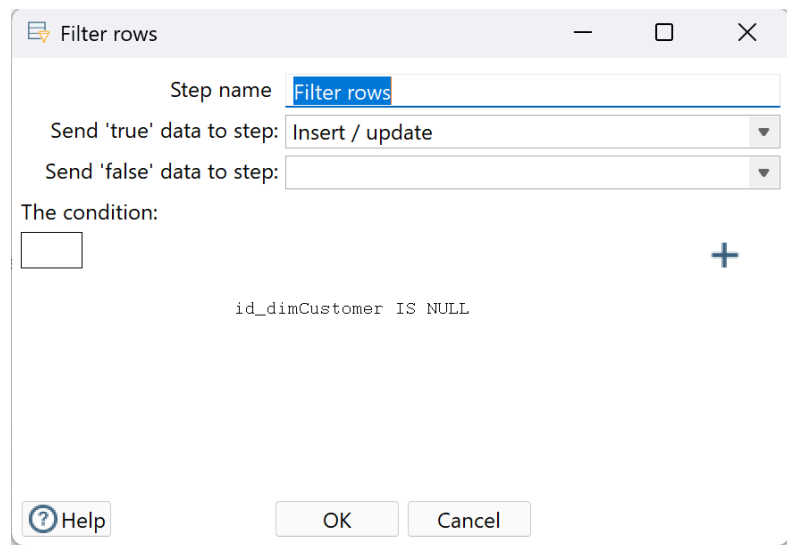
Do not pass the row if the lookup fails ☐

Fail on multiple results? ☐

Order by

[? Help](#) [OK](#) [Cancel](#) [Get Fields](#) [Get lookup fields](#)

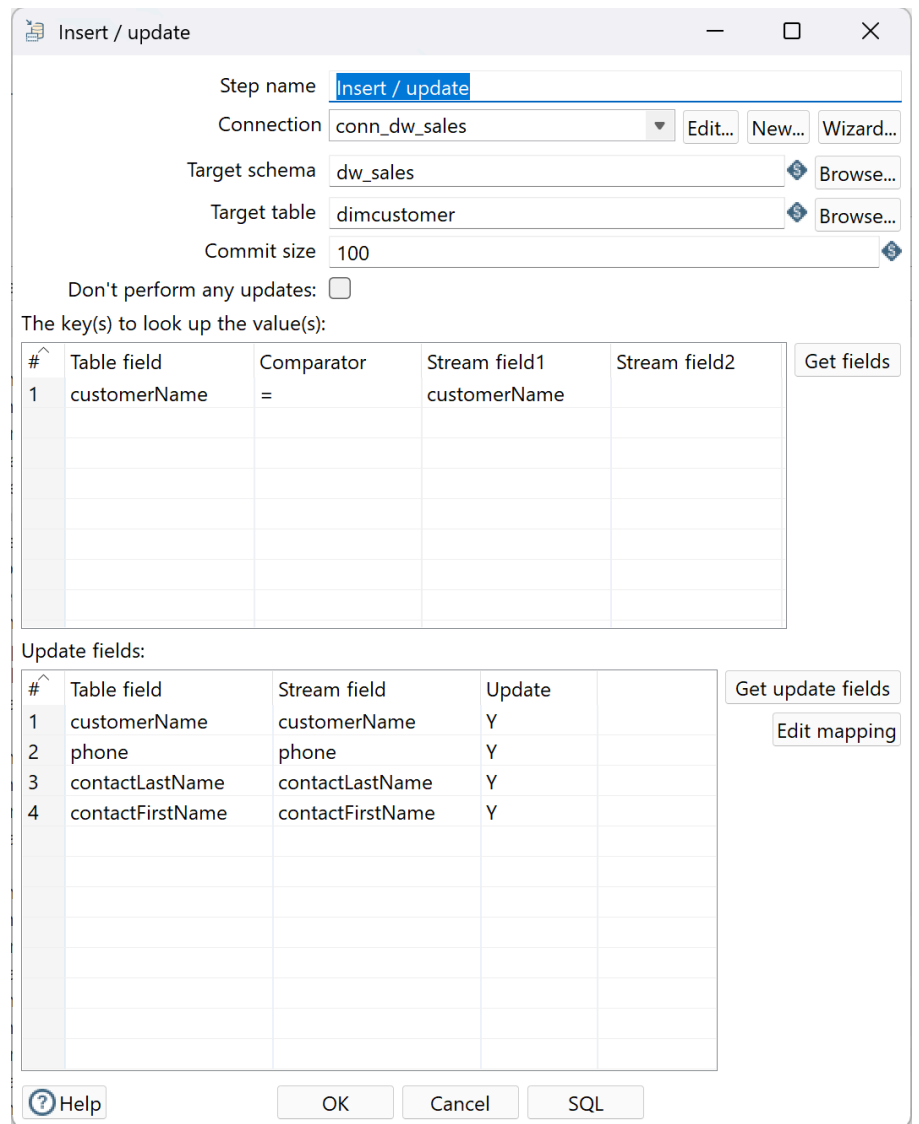
7. Filter rows



The dialog box titled "Filter rows" contains the following fields and controls:

- Step name:** Filter rows
- Send 'true' data to step:** Insert / update
- Send 'false' data to step:** (empty)
- The condition:** id_dimCustomer IS NULL
- Buttons:** Help, OK, Cancel

8. Insert / Update



The dialog box titled "Insert / update" contains the following fields and controls:

- Step name:** Insert / update
- Connection:** conn_dw_sales
- Target schema:** dw_sales
- Target table:** dimcustomer
- Commit size:** 100
- Don't perform any updates:** ☐
- The key(s) to look up the value(s):**

#	Table field	Comparator	Stream field1	Stream field2
1	customerName	=	customerName	
- Update fields:**

#	Table field	Stream field	Update
1	customerName	customerName	Y
2	phone	phone	Y
3	contactLastName	contactLastName	Y
4	contactFirstName	contactFirstName	Y
- Buttons:** Help, OK, Cancel, SQL

9. Hasil run

Diagram showing the data flow steps: CSV file input, Select values, Sort rows, Unique rows, Database lookup, Filter rows, and Insert / update. All steps are marked with a green checkmark.

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

2026-06-03 20:40:54.451 - Spoon - Save file as...
2026-06-03 20:40:54.504 - Spoon - Transformation opened.
2026-06-03 20:40:54.504 - Spoon - Launching transformation [dimcustomer_project uas]...
2026-06-03 20:40:54.504 - Spoon - Started the transformation execution.
2026-06-03 20:40:54.835 - dimcustomer_project uas - Dispatching started for transformation [dimcustomer_project uas]
2026-06-03 20:40:54.886 - CSV file input.0 - Header row skipped in file 'D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv'
2026-06-03 20:40:54.895 - CSV file input.0 - Finished processing (I=2824, O=0, R=0, W=2823, U=0, E=0)
2026-06-03 20:40:54.898 - Select values.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 20:40:54.933 - Sort rows.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 20:40:54.947 - Unique rows.0 - Finished processing (I=0, O=0, R=2823, W=92, U=0, E=0)
2026-06-03 20:40:54.996 - Database lookup.0 - Finished processing (I=0, O=0, R=92, W=92, U=0, E=0)
2026-06-03 20:40:55.009 - Filter rows.0 - Finished processing (I=0, O=0, R=92, W=92, U=0, E=0)
2026-06-03 20:40:55.073 - Insert / update.0 - Finished processing (I=92, O=92, R=92, W=92, U=0, E=0)
2026-06-03 20:40:55.074 - Spoon - The transformation has finished!!

Diagram showing the data flow steps: CSV file input, Select values, Sort rows, Unique rows, Database lookup, Filter rows, and Insert / update. All steps are marked with a green checkmark.

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

#	Stepname	Copied	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	2823	2824	0	0	0	0	0	0.0s	235,333	-
2	Select values	0	2823	2823	0	0	0	0	0	0	0.0s	176,437	-
3	Sort rows	0	2823	2823	0	0	0	0	0	0	0.1s	55,353	-
4	Unique rows	0	2823	92	0	0	0	0	0	0	0.1s	43,431	-
5	Database lookup	0	92	92	0	0	0	0	0	0	0.1s	814	-
6	Filter rows	0	92	92	0	0	0	0	0	0	0.1s	730	-
7	Insert / update	0	92	92	92	92	0	0	0	0	0.2s	482	-

Showing rows 0 - 24 (92 total, Query took 0.0004 seconds.) [id_dimCustomer: 92... - 68...]

SELECT * FROM `dimcustomer` ORDER BY `dimcustomer`.`id\_dimCustomer` DESC

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

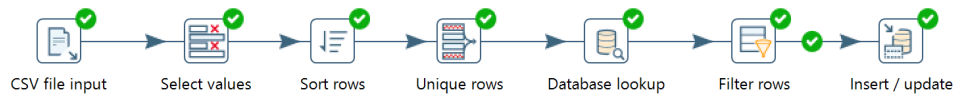
Extra options

	id_dimCustomer	customerName	contactFirstName	contactLastName	phone
☐ Edit ☐ Copy ☐ Delete	92	West Coast Collectables Co.	Steve	Thompson	3105553722
☐ Edit ☐ Copy ☐ Delete	91	Volvo Model Replicas, Co	Christina	Berglund	0921-12 3555
☐ Edit ☐ Copy ☐ Delete	90	Vitachrome Inc.	Michael	Frick	2125551500
☐ Edit ☐ Copy ☐ Delete	89	Vida Sport, Ltd	Michael	Holz	0897-034555
☐ Edit ☐ Copy ☐ Delete	88	UK Collectables, Ltd.	Elizabeth	Devon	(171) 555-2282
☐ Edit ☐ Copy ☐ Delete	87	Toys4GrownUps.com	Julie	Young	6265557265
☐ Edit ☐ Copy ☐ Delete	86	Toys of Finland, Co.	Matti	Karttunen	90-224 8555
☐ Edit ☐ Copy ☐ Delete	85	Toms Speziallitten, Ltd	Henriette	Pfalzheim	0221-5554327
☐ Edit ☐ Copy ☐ Delete	84	Tokyo Collectables, Ltd	Akiko	Shimamura	+81 3 3584 0555
☐ Edit ☐ Copy ☐ Delete	83	The Sharp Gifts Warehouse	Sue	Frick	4085553659
☐ Edit ☐ Copy ☐ Delete	82	Tekni Collectables Inc.	William	Brown	2015559350
☐ Edit ☐ Copy ☐ Delete	81	Technics Stores Inc.	Juri	Hirano	6505556809
☐ Edit ☐ Copy ☐ Delete	80	Super Scale Inc.	Leslie	Murphy	2035559545
☐ Edit ☐ Copy ☐ Delete	79	Suominen Souvenirs	Kalle	Suominen	+358 9 8045 555
☐ Edit ☐ Copy ☐ Delete	78	Stylish Desk Decors, Co.	Ann	Brown	(171) 555-0297
☐ Edit ☐ Copy ☐ Delete	77	Souvenirs And Things Co.	Adrian	Huxley	+61 2 9495 8555
☐ Edit ☐ Copy ☐ Delete	76	Signal Gift Stores	Sue	King	7025551838
☐ Edit ☐ Copy ☐ Delete	75	Signal Collectibles Ltd.	Sue	Taylor	4155554312
☐ Edit ☐ Copy ☐ Delete	74	Scandinavian Gift Ideas	Maria	Larsson	0695-34 6555
☐ Edit ☐ Copy ☐ Delete	73	Saveley & Henriot, Co.	Mary	Saveley	78.32.5555

Console

4. Transformasi dim Date

1. Membuat object



2. Konfigurasi CSV file input

Step name: **CSV file input**

Filename: D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv

Delimiter: ,

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding: mixed

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	ORDERNUMBER	Integer	#	15	0	\$	-	-	none
2	QUANTITYORDERED	Integer	#	15	0	\$	-	-	none
3	PRICEEACH	BigNumber	#.##	5	2	\$	-	-	none
4	ORDERLINENUMBER	Integer	#	15	0	\$	-	-	none
5	SALES	BigNumber	#.##	7	2	\$	-	-	none
6	ORDERDATE	String		15		\$	-	-	none
7	STATUS	String		10		\$	-	-	none
8	QTR_ID	Integer	#	15	0	\$	-	-	none
9	MONTH_ID	Integer	#	15	0	\$	-	-	none
10	YEAR_ID	Integer	#	15	0	\$	-	-	none
11	PRODUCTLINE	String		12		\$	-	-	none
12	MSRP	Integer	#	15	0	\$	-	-	none
13	PRODUCTCODE	String		8		\$	-	-	none
14	CUSTOMERNAME	String		30		\$	-	-	none
15	PHONE	String		16		\$	-	-	none
16	ADDRESSLINE1	String		40		\$	-	-	none
17	ADDRESSLINE2	String		9		\$	-	-	none
18	CITY	String		14		\$	-	-	none
19	STATE	String		10		\$	-	-	none
20	POSTALCODE	String		8		\$	-	-	none
21	COUNTRY	String		9		\$	-	-	none
22	TERRITORY	String		5		\$	-	-	none
23	CONTACTLASTNAME	String		10		\$	-	-	none
24	CONTACTFIRSTNAME	String		9		\$	-	-	none
25	DEALSIZE	String		6		\$	-	-	none

Buttons: OK, Get Fields, Preview, Cancel

3. Select values

- Select & alter

Step name: **Select values**

Tab: Select & Alter

Fields:

#	Fieldname	Rename to	Length	Precision
1	ORDERNUMBER			
2	QUANTITYORDERED			
3	PRICEEACH			
4	ORDERLINENUMBER			
5	SALES			
6	ORDERDATE	date		
7	STATUS			
8	QTR_ID	quarter		
9	MONTH_ID	month		
10	YEAR_ID	year		
11	PRODUCTLINE			
12	MSRP			
13	PRODUCTCODE			
14	CUSTOMERNAME			
15	PHONE			
16	ADDRESSLINE1			
17	ADDRESSLINE2			
18	CITY			
19	STATE			
20	POSTALCODE			
21	COUNTRY			
22	TERRITORY			
23	CONTACTLASTNAME			
24	CONTACTFIRSTNAME			
25	DEALSIZE			

Buttons: Get fields to select, Edit Mapping

Include unspecified fields, ordered by name ☐

Buttons: OK, Cancel

- Remove

Select values

Step nameSelect values

Select & AlterRemoveMeta-data

Fields to remove :

#	Fieldname
1	ORDERNUMBER
2	QUANTITYORDERED
3	PRICEEACH
4	ORDERLINENUMBER
5	SALES
6	STATUS
7	PRODUCTLINE
8	MSRP
9	PRODUCTCODE
10	CUSTOMERNAME
11	PHONE
12	ADDRESSLINE1
13	ADDRESSLINE2
14	CITY
15	STATE
16	POSTALCODE
17	COUNTRY
18	TERRITORY
19	CONTACTLASTNAME
20	CONTACTFIRSTNAME
21	DEALSIZE

Get fields to remove

OKCancel

Help

- Meta data

Select values

Step nameSelect values

Select & AlterRemoveMeta-data

Fields to alter the meta-data for :

#	Fieldname	Rename to	Type	Length	Precision	Binary to Normal?	Format	Date Format Le
1	date		Date			N	M/d/yyyy H:mm	N

Get fields to change

OKCancel

Help

4. Sort rows

[illegible]

5. Unique rows

 Unique rows

— □ ×

Step name Unique rows

Settings

Add counter to output? ☐ Counter field

Redirect duplicate row ☐ Error description 

Fields to compare on (no entries means: compare complete row)

#	Fieldname	Ignore case
1	date	N

 Help

6. Database lookup

Database lookup

Step name

Database lookup

Connection

conn_dw_sales

Edit...

New...

Wizard...

Lookup schema

dw_sales

Browse...

Lookup table

dimdate

Browse...

Enable cache?

☐

Cache size in rows (0=cache)

0

Load all data from table

☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	date	=	date	

Values to return from the lookup table :

#	Field	New name	Default	Type
1	id_dimDate			Integer

Do not pass the row if the lookup fails

☐

Fail on multiple results?

☐

Order by

Help

OK

Cancel

Get Fields

Get lookup fields

7. Filter rows

Filter rows

Step name

Filter rows

Send 'true' data to step:

Insert / update

Send 'false' data to step:

The condition:

+

id_dimDate IS NULL

Help

OK

Cancel

8. Insert / Update

Insert / update

Step name

Insert / update

Connection

conn_dw_sales

Edit...

New...

Wizard...

Target schema

dw_sales

Browse...

Target table

dimdate

Browse...

Commit size

100

Don't perform any updates:

☐

The key(s) to look up the value(s):

#	Table field	Comparator	Stream field1	Stream field2	
1	date	=	date		

Get fields

Update fields:

#	Table field	Stream field	Update	
1	date	date	Y	
2	quarter	quarter	Y	
3	month	month	Y	
4	year	year	Y	

Get update fields

Edit mapping

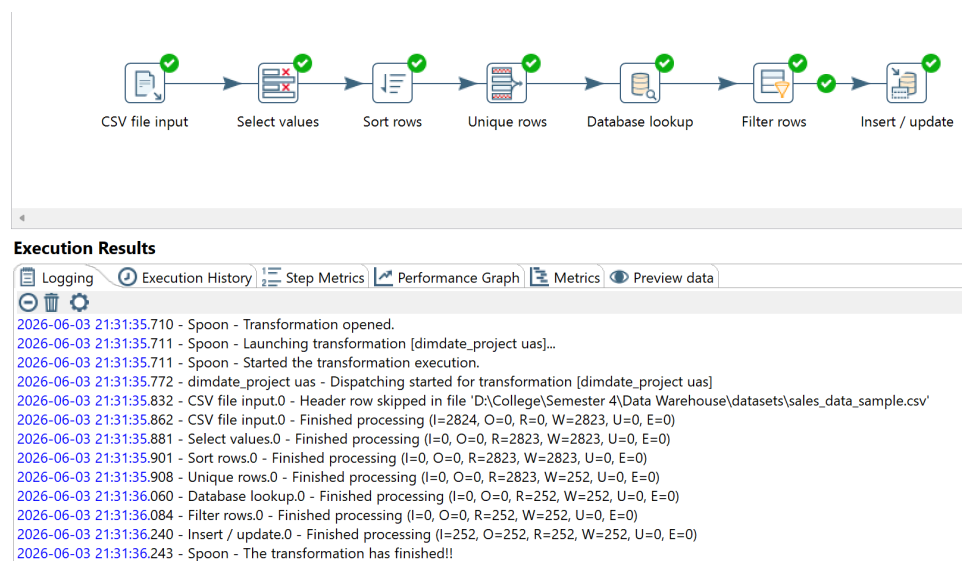
Help

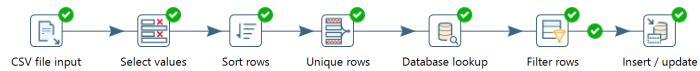
OK

Cancel

SQL

9. Hasil run





Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	2823	2824	0	0	0	0	Finished	0.0s	76,324	-
2	Select values	0	2823	2823	0	0	0	0	0	Finished	0.1s	50,411	-
3	Sort rows	0	2823	2823	0	0	0	0	0	Finished	0.1s	37,145	-
4	Unique rows	0	2823	252	0	0	0	0	0	Finished	0.1s	34,012	-
5	Database lookup	0	252	252	0	0	0	0	0	Finished	0.2s	1,072	-
6	Filter rows	0	252	252	0	0	0	0	0	Finished	0.3s	969	-
7	Insert / update	0	252	252	252	252	0	0	0	Finished	0.4s	606	-

Showing rows 0 - 24 (252 total, Query took 0.0006 seconds.) [id_dimDate: 252... - 228...]

```
SELECT * FROM `dimdate` ORDER BY `dimdate`.`id_dimDate` DESC
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1

>

>>

☐

Show all

Number of rows:

25

Filter rows:

Search

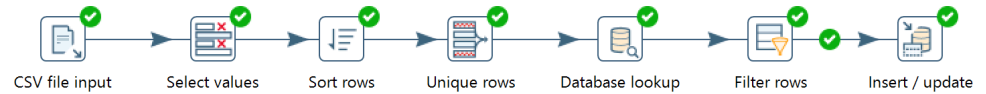
Extra options

				id_dimDate	1	date	month	quarter	year
☐	Edit	Copy	Delete	252	2005-05-31	5	2	2005	
☐	Edit	Copy	Delete	251	2005-05-30	5	2	2005	
☐	Edit	Copy	Delete	250	2005-05-29	5	2	2005	
☐	Edit	Copy	Delete	249	2005-05-17	5	2	2005	
☐	Edit	Copy	Delete	248	2005-05-13	5	2	2005	
☐	Edit	Copy	Delete	247	2005-05-10	5	2	2005	
☐	Edit	Copy	Delete	246	2005-05-09	5	2	2005	
☐	Edit	Copy	Delete	245	2005-05-06	5	2	2005	
☐	Edit	Copy	Delete	244	2005-05-05	5	2	2005	
☐	Edit	Copy	Delete	243	2005-05-03	5	2	2005	
☐	Edit	Copy	Delete	242	2005-05-01	5	2	2005	
☐	Edit	Copy	Delete	241	2005-04-23	4	2	2005	
☐	Edit	Copy	Delete	240	2005-04-22	4	2	2005	
☐	Edit	Copy	Delete	239	2005-04-15	4	2	2005	
☐	Edit	Copy	Delete	238	2005-04-14	4	2	2005	
☐	Edit	Copy	Delete	237	2005-04-08	4	2	2005	
☐	Edit	Copy	Delete	236	2005-04-07	4	2	2005	
☐	Edit	Copy	Delete	235	2005-04-03	4	2	2005	
☐	Edit	Copy	Delete	234	2005-04-01	4	2	2005	
☐	Edit	Copy	Delete	233	2005-03-30	3	1	2005	

Console

5. Transformasi dim Location

1. Membuat object



2. Konfigurasi CSV file input

Step name: **CSV file input**

Filename: D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv

Delimiter: ,

Enclosure: "

NIO buffer size: 50000

Lazy conversion: ☒

Header row present: ☒

Add filename to result: ☐

The row number field name (optional):

Running in parallel: ☐

New line possible in fields: ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	ORDERNUMBER	Integer	#	15	0	\$	.		none
2	QUANTITYORDERED	Integer	#	15	0	\$	.		none
3	PRICEEACH	BigNumber	###	5	2	\$	.		none
4	ORDERLINENUMBER	Integer	#	15	0	\$	.		none
5	SALES	BigNumber	###	7	2	\$	.		none
6	ORDERDATE	String		15		\$	.		none
7	STATUS	String		10		\$	.		none
8	QTR_ID	Integer	#	15	0	\$	.		none
9	MONTH_ID	Integer	#	15	0	\$	.		none
10	YEAR_ID	Integer	#	15	0	\$	.		none
11	PRODUCTLINE	String		12		\$	.		none
12	MSRP	Integer	#	15	0	\$	.		none
13	PRODUCTCODE	String		8		\$	.		none
14	CUSTOMERNAME	String		30		\$	.		none
15	PHONE	String		16		\$	.		none
16	ADDRESSLINE1	String		40		\$	.		none
17	ADDRESSLINE2	String		9		\$	.		none
18	CITY	String		14		\$	.		none
19	STATE	String		10		\$	.		none
20	POSTALCODE	String		8		\$	.		none
21	COUNTRY	String		9		\$	.		none
22	TERRITORY	String		5		\$	.		none
23	CONTACTLASTNAME	String		10		\$	.		none
24	CONTACTFIRSTNAME	String		9		\$	.		none
25	DEALSIZE	String		6		\$	.		none

Buttons: Help, OK, Get Fields, Preview, Cancel

3. Select values

- Select & alter

Step name: **Select values**

Select & Alter | Remove | Meta-data

Fields:

#	Fieldname	Rename to	Length	Precision
1	ORDERNUMBER			
2	QUANTITYORDERED			
3	PRICEEACH			
4	ORDERLINENUMBER			
5	SALES			
6	ORDERDATE			
7	STATUS			
8	QTR_ID			
9	MONTH_ID			
10	YEAR_ID			
11	PRODUCTLINE			
12	MSRP			
13	PRODUCTCODE			
14	CUSTOMERNAME			
15	PHONE			
16	ADDRESSLINE1	addressLine1		
17	ADDRESSLINE2	addressLine2		
18	CITY	city		
19	STATE	state		
20	POSTALCODE	postalCode		
21	COUNTRY	country		
22	TERRITORY	territory		
23	CONTACTLASTNAME			
24	CONTACTFIRSTNAME			
25	DEALSIZE			

Include unspecified fields, ordered by name: ☐

Buttons: Help, OK, Cancel

- Remove

The 'Select values' dialog box is shown with the 'Remove' tab active. The 'Step name' is 'Select values'. The 'Fields to remove' list contains 18 fields: ORDERNUMBER, QUANTITYORDERED, PRICEEACH, ORDERLINENUMBER, SALES, ORDERDATE, STATUS, QTR_ID, MONTH_ID, YEAR_ID, PRODUCTLINE, MSRP, PRODUCTCODE, CUSTOMERNAME, PHONE, CONTACTLASTNAME, CONTACTFIRSTNAME, and DEALSIZE. The 'Get fields to remove' button is on the right. The 'OK' and 'Cancel' buttons are at the bottom.

#	Fieldname
1	ORDERNUMBER
2	QUANTITYORDERED
3	PRICEEACH
4	ORDERLINENUMBER
5	SALES
6	ORDERDATE
7	STATUS
8	QTR_ID
9	MONTH_ID
10	YEAR_ID
11	PRODUCTLINE
12	MSRP
13	PRODUCTCODE
14	CUSTOMERNAME
15	PHONE
16	CONTACTLASTNAME
17	CONTACTFIRSTNAME
18	DEALSIZE

4. Sort rows

The 'Sort rows' dialog box is shown. The 'Step name' is 'Sort rows'. The 'Sort directory' is '%java.io.tmpdir%'. The 'TMP-file prefix' is 'out'. The 'Sort size (rows in memory)' is '1000000'. The 'Free memory threshold (in %)' is empty. The 'Compress TMP Files?' checkbox is checked. The 'Only pass unique rows? (verifies keys only)' checkbox is unchecked. The 'Fields' table has one row: 'city' with 'Ascending' sort order, 'Case sensitive compare?' set to 'N', 'Sort based on current locale?' set to 'N', 'Collator Strength' set to '0', and 'Presorted?' set to 'N'. The 'OK', 'Cancel', and 'Get Fields' buttons are at the bottom.

#	Fieldname	Ascending	Case sensitive compare?	Sort based on current locale?	Collator Strength	Presorted?
1	city	Y	N	N	0	N

5. Unique rows

The 'Unique rows' dialog box is shown. The 'Step name' is 'Unique rows'. The 'Settings' section has 'Add counter to output?' unchecked and 'Redirect duplicate row' unchecked. The 'Fields to compare on (no entries means: compare complete row)' table has one row: 'city' with 'Ignore case' set to 'N'. The 'OK', 'Cancel', and 'Get' buttons are at the bottom.

#	Fieldname	Ignore case
1	city	N

6. Database lookup

The 'Database lookup' window is configured with the following settings:

- Step name: Database lookup
- Connection: conn_dw_sales
- Lookup schema: dw_sales
- Lookup table: dimlocation
- Enable cache?: ☐
- Cache size in rows (0=cache): 0
- Load all data from table: ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	city	=	city	

Values to return from the lookup table :

#	Field	New name	Default	Type
1	id_dimLocation			Integer

Do not pass the row if the lookup fails: ☐
Fail on multiple results?: ☐
Order by:

Buttons: ? Help, OK, Cancel, Get Fields, Get lookup fields

7. Filter rows

The 'Filter rows' window is configured with the following settings:

- Step name: Filter rows
- Send 'true' data to step: Insert / update
- Send 'false' data to step:

The condition:

+

id_dimLocation IS NULL

Buttons: ? Help, OK, Cancel

8. Insert / Update

Insert / update

Step name

Insert / update

Connection

conn_dw_sales

Edit...

New...

Wizard...

Target schema

dw_sales

Browse...

Target table

dimlocation

Browse...

Commit size

100

Don't perform any updates:

☐

The key(s) to look up the value(s):

#	Table field	Comparator	Stream field1	Stream field2
1	city	=	city	

Get fields

Update fields:

#	Table field	Stream field	Update
1	addressLine1	addressLine1	Y
2	addressLine2	addressLine2	Y
3	city	city	Y
4	state	state	Y
5	postalCode	postalCode	Y
6	country	country	Y
7	territory	territory	Y

Get update fields

Edit mapping

Help

OK

Cancel

SQL

9. Hasil run

CSV file input

Select values

Sort rows

Unique rows

Database lookup

Filter rows

Insert / update

Execution Results

Logging

Execution History

Step Metrics

Performance Graph

Metrics

Preview data

2026-06-03 21:49:44.643 - Spoon - Transformation opened.

2026-06-03 21:49:44.644 - Spoon - Launching transformation [dimlocation_project uas]...

2026-06-03 21:49:44.644 - Spoon - Started the transformation execution.

2026-06-03 21:49:44.701 - dimlocation_project uas - Dispatching started for transformation [dimlocation_project uas]

2026-06-03 21:49:44.758 - CSV file input.0 - Header row skipped in file 'D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv'

2026-06-03 21:49:44.767 - CSV file input.0 - Finished processing (I=2824, O=0, R=0, W=2823, U=0, E=0)

2026-06-03 21:49:44.772 - Select values.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)

2026-06-03 21:49:44.783 - Sort rows.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)

2026-06-03 21:49:44.789 - Unique rows.0 - Finished processing (I=0, O=0, R=2823, W=73, U=0, E=0)

2026-06-03 21:49:44.821 - Database lookup.0 - Finished processing (I=0, O=0, R=73, W=73, U=0, E=0)

2026-06-03 21:49:44.822 - Filter rows.0 - Finished processing (I=0, O=0, R=73, W=73, U=0, E=0)

2026-06-03 21:49:44.853 - Insert / update.0 - Finished processing (I=73, O=73, R=73, W=73, U=0, E=0)

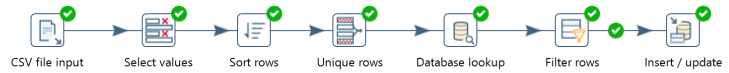
2026-06-03 21:49:44.854 - Spoon - The transformation has finished!!

2026-06-03 22:23:35.719 - Spoon - Transformation opened.

2026-06-03 22:23:35.719 - Spoon - Launching transformation [factorders_project uas]...

2026-06-03 22:23:35.719 - Spoon - Started the transformation execution.

2026-06-03 22:23:38.598 - Spoon - The transformation has finished!!



Execution Results

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	2823	2824	0	0	0	0	Finished	0.0s	235,333	-
2	Select values	0	2823	2823	0	0	0	0	0	Finished	0.0s	166,059	-
3	Sort rows	0	2823	2823	0	0	0	0	0	Finished	0.0s	100,821	-
4	Unique rows	0	2823	73	0	0	0	0	0	Finished	0.0s	80,657	-
5	Database lookup	0	73	73	0	0	0	0	0	Finished	0.1s	1,090	-
6	Filter rows	0	73	73	0	0	0	0	0	Finished	0.1s	1,074	-
7	Insert / update	0	73	73	73	73	0	0	0	Finished	0.1s	745	-

Showing rows 0 - 24 (73 total, Query took 0.0005 seconds.) [id_dimLocation: 73... - 49...]

`SELECT * FROM 'dimlocation' ORDER BY 'dimlocation`.`id_dimLocation` DESC`

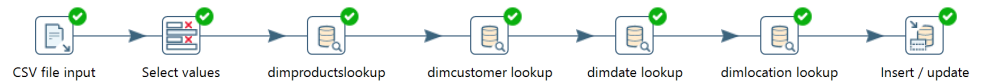
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

		id_dimLocation	addressLine1	addressLine2	city	state	postalCode	country	territory
☐	Edit Copy Delete	73	3758 North Pendale Street	NULL	White Plains	NY	24067	USA	NA
☐	Edit Copy Delete	72	67, avenue de l'Europe	NULL	Versailles	NULL	78000	France	EMEA
☐	Edit Copy Delete	71	1900 Oak St.	NULL	Vancouver	BC	V3F 2K1	Canada	NA
☐	Edit Copy Delete	70	23 Tsawassen Blvd.	NULL	Tsawassen	BC	T2F 8M4	Canada	NA
☐	Edit Copy Delete	69	1 rue Alsace-Lorraine	NULL	Toulouse	NULL	31000	France	EMEA
☐	Edit Copy Delete	68	Via Monte Bianco 34	NULL	Torino	NULL	10100	Italy	EMEA
☐	Edit Copy Delete	67	24, place Kluber	NULL	Strasbourg	NULL	67000	France	EMEA
☐	Edit Copy Delete	66	Erling Skakkes gate 78	NULL	Stavern	NULL	4110	Norway	EMEA
☐	Edit Copy Delete	65	31 Duncan St. West End	NULL	South Brisbane	Queensland	4101	Australia	APAC
☐	Edit Copy Delete	64	Bronz Sok., Bronz Apt. 3/6 Tesvikiye	NULL	Singapore	NULL	79903	Singapore	Japan
☐	Edit Copy Delete	63	C/ Romero, 33	NULL	Sevilla	NULL	41101	Spain	EMEA
☐	Edit Copy Delete	62	5677 Strong St.	NULL	San Rafael	CA	97562	USA	NA
☐	Edit Copy Delete	61	3086 Ingle Ln.	NULL	San Jose	CA	94217	USA	NA
☐	Edit Copy Delete	60	7734 Strong St.	NULL	San Francisco	CA	NULL	USA	NA
☐	Edit Copy Delete	59	361 Furth Circle	NULL	San Diego	CA	91217	USA	NA
☐	Edit Copy Delete	58	Geislweg 14	NULL	Salzburg	NULL	5020	Austria	EMEA
☐	Edit Copy Delete	57	59 rue de l'Abbaye	NULL	Reims	NULL	51100	France	EMEA
☐	Edit Copy Delete	56	Strada Provinciale 124	NULL	Reggio Emilia	NULL	42100	Italy	EMEA
☐	Edit Copy Delete	55	782 First Street	NULL	Philadelphia	PA	71270	USA	NA
☐	Edit Copy Delete	54	78934 Hillside Dr.	NULL	Pasadena	CA	90003	USA	NA

6. Transformasi Facts_orders(Tabel Fakta)

1. Membuat object



2. Konfigurasi CSV file input

Step name: **CSV file input**

Filename: D:\College\Semester 4\Data Warehouse\dataset\sales_data_sample.csv

Delimiter: ,

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result: ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	ORDERNUMBER	Integer	#	15	0	\$	-	-	none
2	QUANTITYORDERED	Integer	#	15	0	\$	-	-	none
3	PRICEEACH	BigNumber	#,##	5	2	\$	-	-	none
4	ORDERLINENUMBER	Integer	#	15	0	\$	-	-	none
5	SALES	BigNumber	#,##	7	2	\$	-	-	none
6	ORDERDATE	String		15		\$	-	-	none
7	STATUS	String		10		\$	-	-	none
8	QTR_ID	Integer	#	15	0	\$	-	-	none
9	MONTH_ID	Integer	#	15	0	\$	-	-	none
10	YEAR_ID	Integer	#	15	0	\$	-	-	none
11	PRODUCTLINE	String		16		\$	-	-	none
12	MSRP	Integer	#	15	0	\$	-	-	none
13	PRODUCTCODE	String		8		\$	-	-	none
14	CUSTOMERNAME	String		30		\$	-	-	none
15	PHONE	String		16		\$	-	-	none
16	ADDRESSLINE1	String		40		\$	-	-	none
17	ADDRESSLINE2	String		9		\$	-	-	none
18	CITY	String		14		\$	-	-	none
19	STATE	String		10		\$	-	-	none
20	POSTALCODE	String		8		\$	-	-	none
21	COUNTRY	String		9		\$	-	-	none
22	TERRITORY	String		5		\$	-	-	none
23	CONTACTLASTNAME	String		10		\$	-	-	none
24	CONTACTFIRSTNAME	String		9		\$	-	-	none
25	DEALSIZE	String		6		\$	-	-	none

Help OK Get Fields Preview Cancel

3. Select values

- Select & alter

Step name: **Select values**

Select & Alter Remove Meta-data

Fields :

#	Fieldname	Rename to	Length	Precision
1	ORDERNUMBER	orderNumber		
2	QUANTITYORDERED	quantityOrdered		
3	PRICEEACH	priceEach		
4	ORDERLINENUMBER	orderLineNumber		
5	SALES	sales		
6	ORDERDATE	date		
7	STATUS	status		
8	QTR_ID			
9	MONTH_ID			
10	YEAR_ID			
11	PRODUCTLINE			
12	MSRP			
13	PRODUCTCODE	productCode		
14	CUSTOMERNAME	customerName		
15	PHONE			
16	ADDRESSLINE1			
17	ADDRESSLINE2			
18	CITY	city		
19	STATE			
20	POSTALCODE			
21	COUNTRY			
22	TERRITORY			
23	CONTACTLASTNAME			
24	CONTACTFIRSTNAME			
25	DEALSIZE	dealSize		

Get fields to select
Edit Mapping

Include unspecified fields, ordered by name ☐

Help OK Cancel

- Remove

Step name: **Select values**

Select & Alter Remove Meta-data

Fields to remove :

#	Fieldname
1	QTR_ID
2	MONTH_ID
3	YEAR_ID
4	PRODUCTLINE
5	MSRP
6	PHONE
7	ADDRESSLINE1
8	ADDRESSLINE2
9	STATE
10	POSTALCODE
11	COUNTRY
12	TERRITORY
13	CONTACTLASTNAME
14	CONTACTFIRSTNAME

Get fields to remove

Help OK Cancel

- Meta data

Select values

Step name: **Select values**

Select & Alter Remove Meta-data

Fields to alter the meta-data for:

#	Fieldname	Rename to	Type	Length	Precision	Binary to Normal?	Format	Date Format Le
1	date		Date			N	M/d/yyyy H:mm	N

Get fields to change

OK Cancel

Help

4. dimproductslookup

Database lookup

Step name

dimproductslookup

Connection

conn_dw_sales

Edit...

New...

Wizard...

Lookup schema

dw_sales

Browse...

Lookup table

dimproducts

Browse...

Enable cache?

☐

Cache size in rows (0=cache)

0

Load all data from table

☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	productCode	=	productCode	

Values to return from the lookup table :

#	Field	New name	Default	Type
1	id_dimProduct			Integer

Do not pass the row if the lookup fails

☐

Fail on multiple results?

☐

Order by

Help

OK

Cancel

Get Fields

Get lookup fields

5. dimcustomerlookup

Database lookup

Step name: dimcustomer lookup

Connection: conn_dw_sales Edit... New... Wizard...

Lookup schema: dw_sales Browse...

Lookup table: dimcustomer Browse...

Enable cache? ☐

Cache size in rows (0=cache): 0

Load all data from table ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	customerName	=	customerName	

Values to return from the lookup table :

#	Field	New name	Default	Type
1	id_dimCustomer			Integer

Do not pass the row if the lookup fails ☐

Fail on multiple results? ☐

Order by:

Help OK Cancel Get Fields Get lookup fields

6. dimdatelookup

Database lookup

Step name: dimdate lookup

Connection: conn_dw_sales Edit... New... Wizard...

Lookup schema: dw_sales Browse...

Lookup table: dimdate Browse...

Enable cache? ☐

Cache size in rows (0=cache): 0

Load all data from table ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	date	=	date	

Values to return from the lookup table :

#	Field	New name	Default	Type
1	id_dimDate			Integer

Do not pass the row if the lookup fails ☐

Fail on multiple results? ☐

Order by:

Help OK Cancel Get Fields Get lookup fields

7. dimlocationlookup

Database lookup

Step name: dimlocation lookup

Connection: conn_dw_sales Edit... New... Wizard...

Lookup schema: dw_sales Browse...

Lookup table: dimlocation Browse...

Enable cache? ☐

Cache size in rows (0=cache): 0

Load all data from table ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	city	=	city	

Values to return from the lookup table:

#	Field	New name	Default	Type
1	id_dimLocation			Integer

Do not pass the row if the lookup fails ☐

Fail on multiple results? ☐

Order by:

Help OK Cancel Get Fields Get lookup fields

8. Insert / Update

Insert / update

Step name: insert / update

Connection: conn_dw_sales Edit... New... Wizard...

Target schema: dw_sales Browse...

Target table: fact_orders Browse...

Commit size: 100

Don't perform any updates: ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Stream field1	Stream field2
1	orderNumber	=	orderNumber	
2	orderLineNumber	=	orderLineNumber	

Get fields

Update fields:

#	Table field	Stream field	Update
1	orderNumber	orderNumber	Y
2	quantityOrdered	quantityOrdered	Y
3	priceEach	priceEach	Y
4	orderLineNumber	orderLineNumber	Y
5	sales	sales	Y
6	status	status	Y
7	dealSize	dealSize	Y
8	id_dimProduct	id_dimProduct	Y
9	id_dimCustomer	id_dimCustomer	Y
10	id_dimDate	id_dimDate	Y
11	id_dimLocation	id_dimLocation	Y

Get update fields

Edit mapping

Help OK Cancel SQL

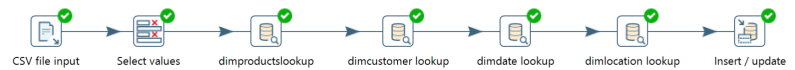
9. Hasil run



Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

2026-06-03 22:23:35.719 - Spoon - Transformation opened.
2026-06-03 22:23:35.719 - Spoon - Launching transformation [factorders_project uas]..
2026-06-03 22:23:35.719 - Spoon - Started the transformation execution.
2026-06-03 22:23:35.750 - factorders_project uas - Dispatching started for transformation [factorders_project uas]
2026-06-03 22:23:35.830 - CSV file input.0 - Header row skipped in file 'D:\College\Semester 4\Data Warehouse\datasets\sales_data_sample.csv'
2026-06-03 22:23:35.838 - CSV file input.0 - Finished processing (I=2824, O=0, R=0, W=2823, U=0, E=0)
2026-06-03 22:23:35.844 - Select values.0 - Finished processing (I=0, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:36.840 - dimproductslookup.0 - Finished processing (I=2823, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:36.887 - dimcustomer lookup.0 - Finished processing (I=2823, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:36.951 - dimdate lookup.0 - Finished processing (I=2823, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:37.000 - dimlocation lookup.0 - Finished processing (I=2823, O=0, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:38.596 - Insert / update.0 - Finished processing (I=2823, O=2823, R=2823, W=2823, U=0, E=0)
2026-06-03 22:23:38.598 - Spoon - The transformation has finished!!



Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	2823	2824	0	0	0	0	Finished	0.0s	282,400	-
2	Select values	0	2823	2823	0	0	0	0	0	Finished	0.0s	188,200	-
3	dimproductslookup	0	2823	2823	2823	0	0	0	0	Finished	1.0s	2,790	-
4	dimcustomer lookup	0	2823	2823	2823	0	0	0	0	Finished	1.1s	2,666	-
5	dimdate lookup	0	2823	2823	2823	0	0	0	0	Finished	1.1s	2,514	-
6	dimlocation lookup	0	2823	2823	2823	0	0	0	0	Finished	1.2s	2,409	-
7	Insert / update	0	2823	2823	2823	2823	0	0	0	Finished	2.8s	1,020	-

Showing rows 0 - 24 (2823 total, Query took 0.0004 seconds.) [id_factOrder: 2823... - 2799...]

SELECT * FROM 'fact_orders' ORDER BY 'fact_orders'. 'id_factOrder' DESC

Profiling Edit inline Edit Explain SQL Create PHP code Refresh

1 > >> Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

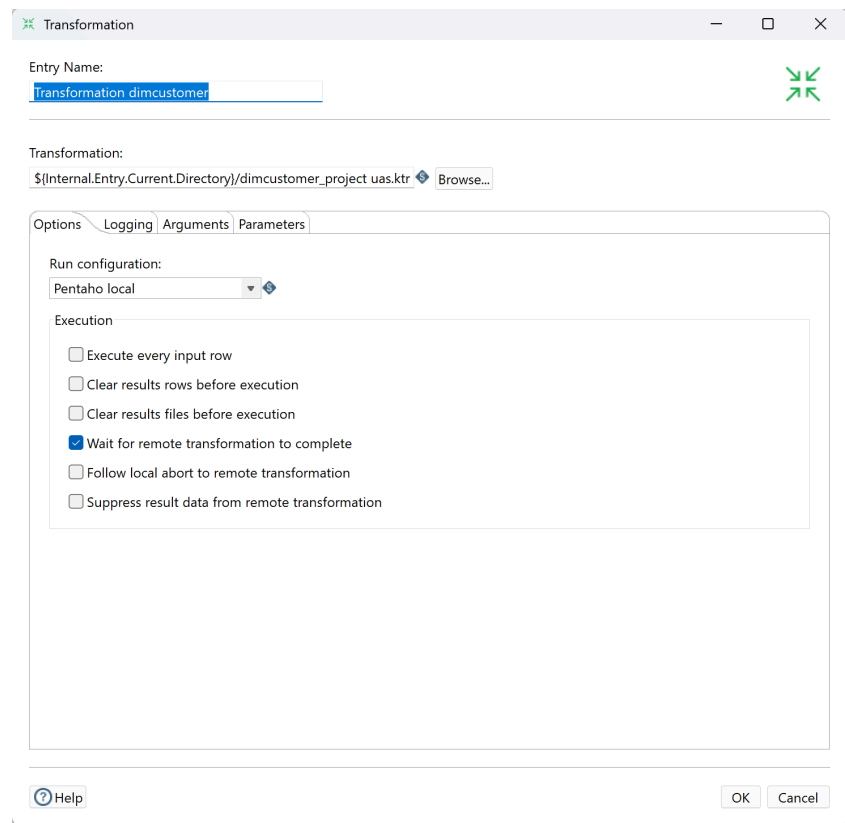
	id_factOrder	orderNumber	id_dimProduct	id_dimCustomer	id_dimDate	id_dimLocation	quantityOrdered	priceEach	orderLineNumber	sales	status	dealSize
☐	2823	10414	109	38	245	7	47	65.52	9	3079.44	On Hold	Medium
☐	2822	10397	109	1	232	69	34	62.24	1	2116.16	Shipped	Small
☐	2821	10386	109	34	223	35	43	100.00	4	5417.57	Resolved	Medium
☐	2820	10373	109	65	212	52	29	100.00	1	3978.51	Shipped	Medium
☐	2819	10350	109	34	195	35	20	100.00	15	2244.40	Shipped	Small
☐	2818	10337	109	20	188	49	42	97.16	5	4080.72	Shipped	Medium
☐	2817	10327	109	28	180	26	37	86.74	4	3209.38	Resolved	Medium
☐	2816	10315	109	46	173	43	40	55.69	5	2227.60	Shipped	Small
☐	2815	10306	109	11	168	37	35	59.51	6	2082.85	Shipped	Small
☐	2814	10293	109	2	159	68	32	60.06	1	1921.92	Shipped	Small
☐	2813	10283	109	70	150	70	33	51.32	12	1693.56	Shipped	Small
☐	2812	10273	109	66	142	11	37	45.86	10	1696.82	Shipped	Small
☐	2811	10261	109	67	132	41	29	50.78	7	1472.62	Shipped	Small
☐	2810	10248	109	47	122	49	23	65.52	9	1506.96	Cancelled	Small
☐	2809	10232	109	39	109	17	24	49.69	3	1192.56	Shipped	Small
☐	2808	10222	109	22	99	59	36	63.34	18	2280.24	Shipped	Small
☐	2807	10208	109	73	86	34	42	63.88	6	2682.96	Shipped	Small
☐	2806	10197	109	33	79	3	42	50.23	12	2109.66	Shipped	Small
☐	2805	10185	109	56	73	45	28	64.43	6	1804.04	Shipped	Small
☐	2804	10177	109	16	68	35	40	50.23	6	2009.20	Shipped	Small

7. Job

1. Membuat object



2. Konfigurasi transformation dimcustomer

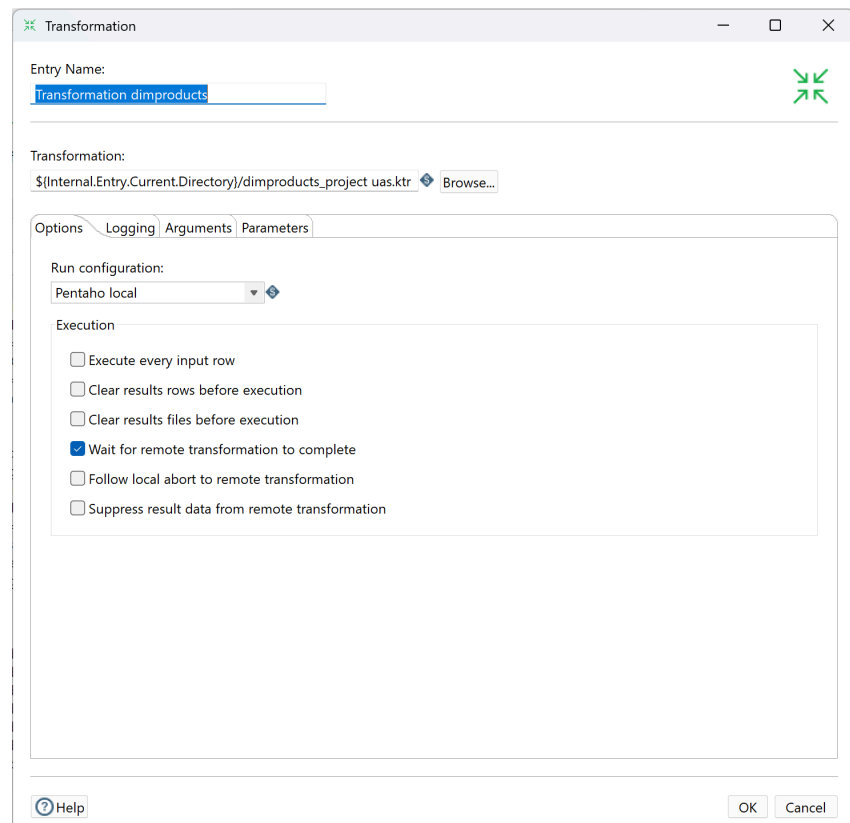


The screenshot shows the 'Transformation' configuration window for 'Transformation dimcustomer'. The 'Entry Name' field is set to 'Transformation dimcustomer'. The 'Transformation' field contains the path '\$(Internal.Entry.Current.Directory)/dimcustomer_project uas.ktr' with a 'Browse...' button. The 'Options' tab is selected, showing the 'Run configuration' dropdown set to 'Pentaho local'. Under the 'Execution' section, the following options are listed:

- ☐ Execute every input row
- ☐ Clear results rows before execution
- ☐ Clear results files before execution
- ☒ Wait for remote transformation to complete
- ☐ Follow local abort to remote transformation
- ☐ Suppress result data from remote transformation

At the bottom, there is a 'Help' button and 'OK' and 'Cancel' buttons.

3. Konfigurasi transformation dimproducts

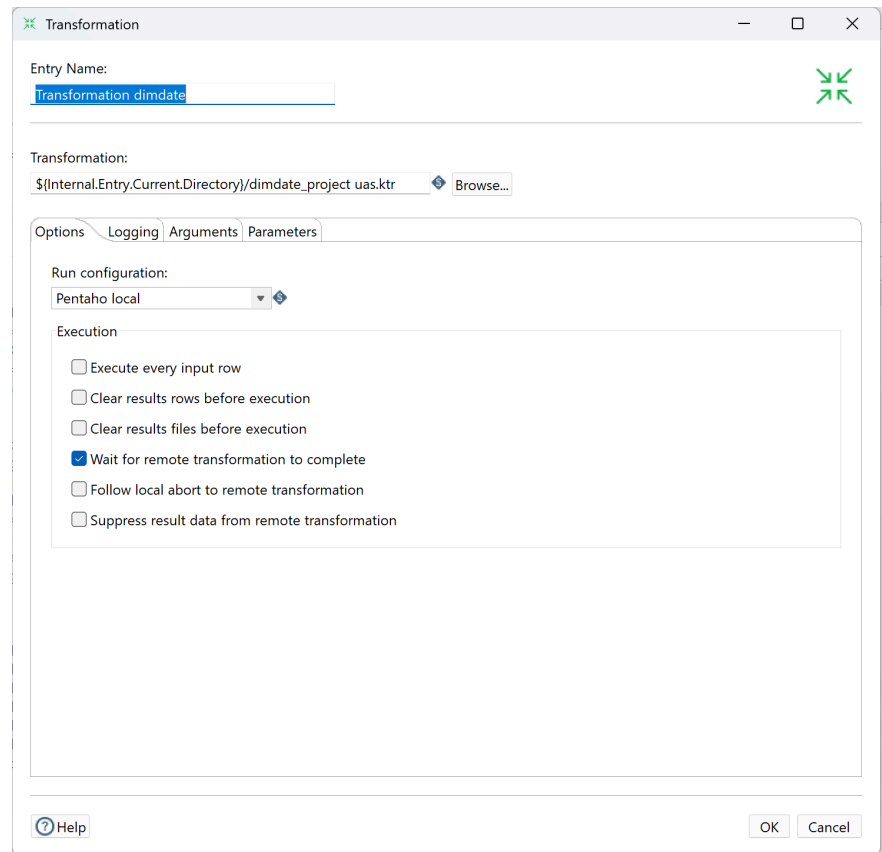


The screenshot shows the 'Transformation' configuration window for 'Transformation dimproducts'. The 'Entry Name' field is set to 'Transformation dimproducts'. The 'Transformation' field contains the path '\$(Internal.Entry.Current.Directory)/dimproducts_project uas.ktr' with a 'Browse...' button. The 'Options' tab is selected, showing the 'Run configuration' dropdown set to 'Pentaho local'. Under the 'Execution' section, the following options are listed:

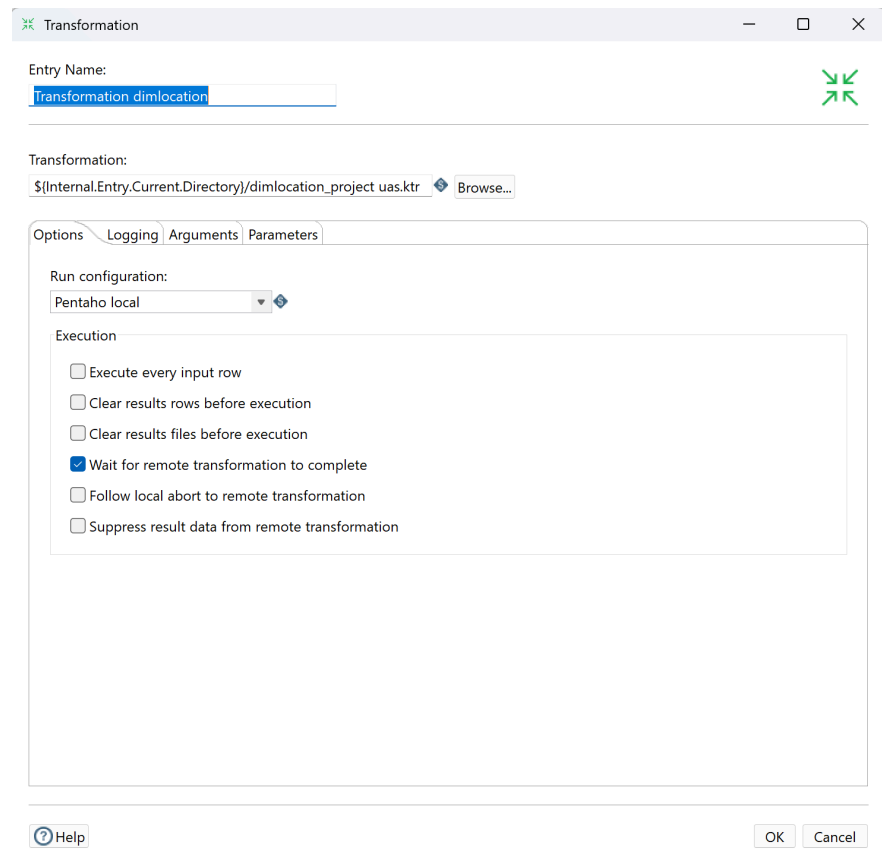
- ☐ Execute every input row
- ☐ Clear results rows before execution
- ☐ Clear results files before execution
- ☒ Wait for remote transformation to complete
- ☐ Follow local abort to remote transformation
- ☐ Suppress result data from remote transformation

At the bottom, there is a 'Help' button and 'OK' and 'Cancel' buttons.

4. Konfigurasi transformation dimdate



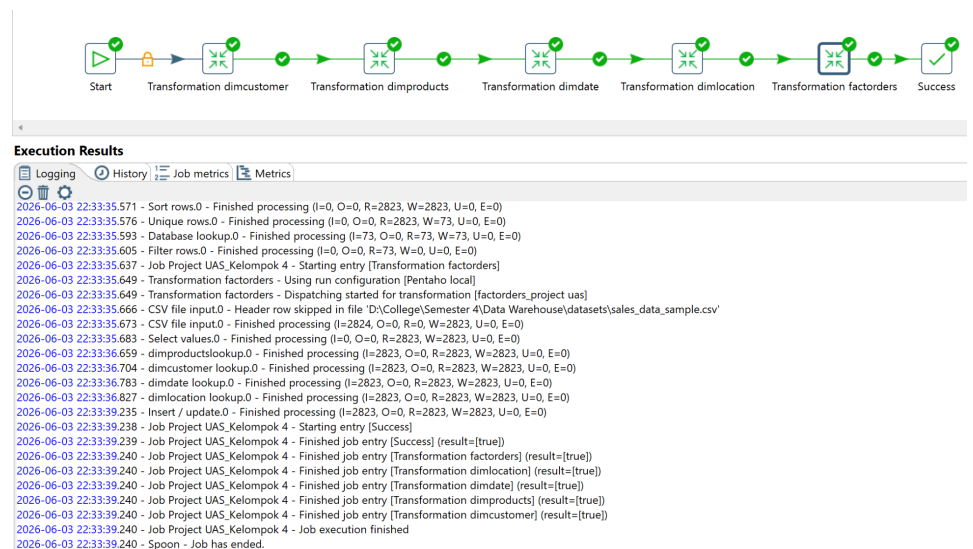
5. Konfigurasi transformation dimlocation

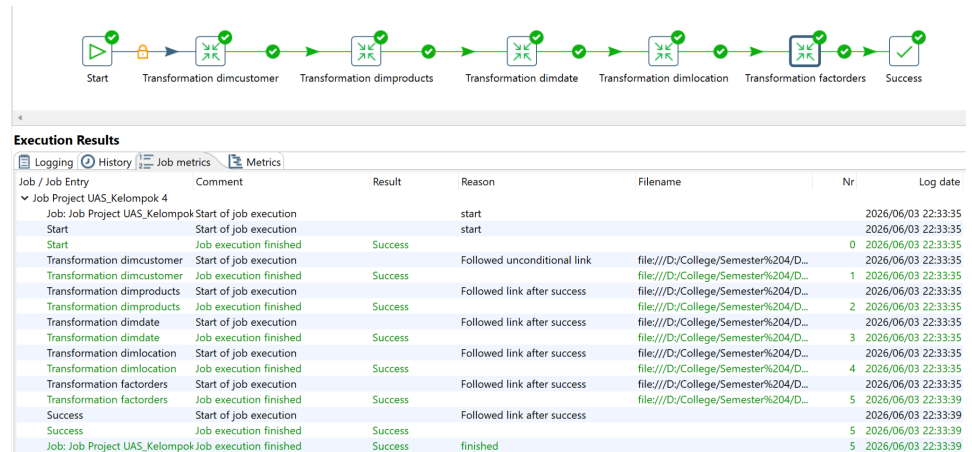


6. Konfigurasi transformation factorders

The screenshot shows the 'Transformation' configuration window in Pentaho. The 'Entry Name' is 'Transformation factorders'. The 'Transformation' path is set to '\$[Internal.Entry.Current.Directory]/factorders_project uas.ktr'. The 'Options' tab is selected, showing the 'Run configuration' as 'Pentaho local'. Under the 'Execution' section, the following options are checked: 'Wait for remote transformation to complete'. Other options like 'Execute every input row', 'Clear results rows before execution', 'Clear results files before execution', 'Follow local abort to remote transformation', and 'Suppress result data from remote transformation' are unchecked. The window has 'Help', 'OK', and 'Cancel' buttons at the bottom.

7. Hasil run





E. Tahap Load (Pemuatan Data)

Setelah proses transformasi selesai, data dimuat ke dalam database dw_sales menggunakan step Insert/Update di Pentaho Spoon. Proses load dilakukan dengan memperhatikan integritas data, kunci primer, dan relasi antar tabel. Tabel dimensi dimuat terlebih dahulu sebelum tabel fakta untuk memastikan seluruh foreign key pada fact_orders sudah tersedia di tabel dimensi yang bersesuaian, sehingga tidak terjadi pelanggaran referential integrity. Setiap tabel dimensi menggunakan primary key AUTO_INCREMENT yang di-generate otomatis oleh MySQL, sedangkan tabel fakta menggunakan foreign key hasil lookup dari masing-masing dimensi.

1. Load dimcustomer

Data pelanggan berhasil dimuat sebanyak 92 baris ke tabel dimcustomer. Proses ini melibatkan penghapusan duplikat berdasarkan customerName sehingga setiap pelanggan hanya tersimpan satu kali. Tidak ditemukan error pada proses load ini.

2. Load dimproducts

Data produk berhasil dimuat sebanyak 109 baris ke tabel dimproducts. Duplikat berdasarkan productCode berhasil dieliminasi sehingga setiap produk hanya tercatat satu kali dalam dimensi produk.

3. Load dimdate

Data waktu berhasil dimuat sebanyak 252 baris ke tabel dimdate. Kolom date berhasil dikonversi dari format String ke tipe Date dengan format M/d/yyyy H:mm sebelum dimuat ke database.

4. Load dimlocation

Data lokasi berhasil dimuat sebanyak 73 baris ke tabel dimlocation. Duplikat berdasarkan city berhasil dieliminasi sehingga setiap lokasi hanya tersimpan satu kali.

5. Load fact_orders

Data transaksi berhasil dimuat sebanyak 2.823 baris ke tabel fact_orders. Proses ini melibatkan lookup ke seluruh tabel dimensi untuk mendapatkan

foreign key yang sesuai, yaitu id_dimProduct, id_dimCustomer, id_dimDate, dan id_dimLocation, sebelum data dimasukkan ke tabel fakta.

6. Eksekusi Job

Seluruh transformation dijalankan secara berurutan melalui satu Job di Pentaho Spoon. Job berhasil dieksekusi tanpa error dengan urutan: dimcustomer_project uas → dimproducts_project uas → dimdate_project uas → dimlocation_project uas → factorders_project uas. Seluruh data dari file CSV sales_data_sample.csv yang terdiri dari 2.823 baris transaksi berhasil dimuat ke dalam data warehouse dw_sales dengan struktur star schema yang telah dirancang.

IV. HASIL DAN ANALISIS KEY PERFORMANCE INDICATORS (KPI)

A. Hasil Implementasi Data Warehouse

Setelah proses ETL selesai dijalankan, data warehouse GlobalScale Retail berhasil dibangun dengan struktur star schema yang terdiri dari satu tabel fakta dan empat tabel dimensi. Data yang semula tersebar dalam satu file CSV flat berhasil dinormalisasi ke dalam struktur relasional yang terpisah, siap untuk digunakan dalam analisis multidimensional.

Keberhasilan implementasi ditunjukkan dengan terpenuhinya integritas referensial antar tabel, tidak adanya nilai null pada kolom kunci, serta konsistensi jumlah record antara sumber data dan data warehouse. Dari dataset awal yang memiliki 2.823 baris transaksi, seluruh data berhasil dimuat ke dalam tabel fakta tanpa kehilangan data.

B. Analisis Key Performance Indicators (KPI)

Data warehouse yang telah dibangun dapat dimanfaatkan untuk menghasilkan berbagai laporan analitik bisnis. Berikut adalah tiga KPI utama yang dapat dihasilkan beserta contoh query SQL yang digunakan:

KPI 1: Total Revenue per Tahun

KPI ini mengukur total pendapatan (revenue) perusahaan secara keseluruhan di setiap tahun. Dari data terlihat bahwa 2004 adalah tahun dengan revenue tertinggi (\$4,72 juta), naik sekitar 34% dari 2003. Revenue 2005 tampak rendah (\$1,79 juta) karena data hanya mencakup sampai pertengahan 2005 (tidak full year). Tren 2003→2004 menunjukkan pertumbuhan bisnis yang signifikan.

Query :

```
SELECT
```

```
    d.year AS tahun,
```

```

SUM(f.sales) AS total_revenue,

COUNT(f.id_factOrder) AS jumlah_transaksi

FROM fact_orders f

JOIN dimdate d ON f.id_dimDate = d.id_dimDate

GROUP BY d.year

ORDER BY d.year;

```

Hasil :

✓ Showing rows 0 - 2 (3 total, Query took 0.0003 seconds.)

```

SELECT d.year AS tahun, SUM(f.sales) AS total_revenue, COUNT(f.id_factOrder) AS jumlah_transaksi FROM fact_orders f JOIN dimdate d ON f.id_dimDate = d.id_dimDate GROUP BY d.year ORDER BY d.year;

```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 ▾ | Filter rows:

Extra options

tahun	total_revenue	jumlah_transaksi
2003	3516979.54	1000
2004	4724162.60	1345
2005	1791486.71	478

KPI 2: Product Line dengan Revenue Tertinggi

KPI ini mengidentifikasi lini produk mana yang paling menguntungkan. Classic Cars mendominasi dengan revenue \$3,9 juta — hampir 2x lipat dibanding Vintage Cars di posisi kedua. Sementara Trains adalah kontributor terkecil dengan hanya \$226K. Insight ini berguna untuk keputusan manajemen seperti alokasi stok, strategi promosi, dan prioritas pengadaan produk.

Query :

```

SELECT

p.productLine AS kategori_produk,

SUM(f.sales) AS total_revenue,

SUM(f.quantityOrdered) AS total_unit_terjual,

ROUND(AVG(f.priceEach), 2) AS rata_rata_harga

FROM fact_orders f

JOIN dimproducts p ON f.id_dimProduct = p.id_dimProduct

GROUP BY p.productLine

ORDER BY total_revenue DESC;

```

Hasil :

Showing rows 0 - 6 (7 total, Query took 0.0092 seconds.)

```
SELECT p.productLine AS kategori_produk, SUM(f.sales) AS total_revenue, SUM(f.quantityOrdered) AS total_unit_terjual, ROUND(AVG(f.priceEach), 2) AS rata_rata_harga FROM fact_orders f JOIN dimproducts p ON f.id_dimProduct = p.id_dimProduct GROUP BY p.productLine ORDER BY total_revenue DESC;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

kategori_produk	total_revenue	total_unit_terjual	rata_rata_harga
Classic Cars	3919615.66	33992	87.34
Vintage Cars	1903150.84	21069	78.15
Motorcycles	1166388.34	11663	83.00
Trucks and Buses	1127789.84	10777	87.53
Planes	975003.57	10727	81.74
Ships	714437.13	8127	83.86
Trains	226243.47	2712	75.65

KPI 3: Deal Size yang Paling Banyak Menghasilkan Revenue

KPI ini menganalisis segmentasi transaksi berdasarkan ukuran deal. Menariknya, Medium Deal adalah kontributor terbesar (\$6,08 juta / ~61% dari total revenue), bukan Large Deal. Hal ini menunjukkan bahwa bisnis perusahaan lebih banyak ditopang oleh transaksi menengah yang konsisten dan banyak jumlahnya, bukan dari sedikit transaksi bernilai besar. Ini penting untuk strategi penetapan harga dan segmentasi pelanggan.

Query :

```
SELECT

    f.dealSize AS ukuran_deal,

    COUNT(f.id_factOrder) AS jumlah_order,

    SUM(f.sales) AS total_revenue,

    ROUND(AVG(f.sales), 2) AS rata_rata_nilai_order

FROM fact_orders f

GROUP BY f.dealSize

ORDER BY total_revenue DESC;
```

Hasil :

Showing rows 0 - 2 (3 total, Query took 0.0039 seconds.)

```
SELECT f.dealSize AS ukuran_deal, COUNT(f.id_factOrder) AS jumlah_order, SUM(f.sales) AS total_revenue, ROUND(AVG(f.sales), 2) AS rata_rata_nilai_order FROM fact_orders f GROUP BY f.dealSize ORDER BY total_revenue DESC;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

ukuran_deal	jumlah_order	total_revenue	rata_rata_nilai_order
Medium	1384	6087432.24	4398.43
Small	1282	2643077.35	2061.68
Large	157	1302119.26	8293.75

KPI 4: Top 10 Pelanggan dengan Pembelian Terbanyak

KPI ini mengidentifikasi pelanggan paling bernilai (high-value customers). Euro Shopping Channel adalah pelanggan terbesar dengan \$912K — hampir 1,4x lipat dari pelanggan kedua. Dua pelanggan teratas (Euro Shopping Channel + Mini Gifts Distributors) menyumbang sekitar 16% dari total revenue keseluruhan, menunjukkan ketergantungan yang cukup tinggi pada pelanggan kunci. Perusahaan perlu menjaga loyalitas kedua pelanggan ini sebagai prioritas.

Query :

```
SELECT

    c.customerName AS nama_pelanggan,

    COUNT(f.id_factOrder) AS jumlah_transaksi,

    SUM(f.quantityOrdered) AS total_unit_dibeli,

    SUM(f.sales) AS total_belanja

FROM fact_orders f

JOIN dimcustomer c ON f.id_dimCustomer = c.id_dimCustomer

GROUP BY c.customerName

ORDER BY total_belanja DESC

LIMIT 10;
```

Hasil :

Showing rows 0 - 9 (10 total, Query took 0.0071 seconds.)

```
SELECT c.customerName AS nama_pelanggan, COUNT(f.id_factOrder) AS jumlah_transaksi, SUM(f.quantityOrdered) AS total_unit_dibeli, SUM(f.sales) AS total_belanja
FROM fact_orders f JOIN dimcustomer c ON f.id_dimCustomer = c.id_dimCustomer GROUP BY c.customerName ORDER BY total_belanja DESC LIMIT 10;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

Extra options

nama_pelanggan	jumlah_transaksi	total_unit_dibeli	total_belanja
Euro Shopping Channel	259	9327	912294.11
Mini Gifts Distributors Ltd.	180	6366	654858.06
Australian Collectors, Co.	55	1926	200995.41
Muscle Machine Inc	48	1775	197736.94
La Rochelle Gifts	53	1832	180124.90
Dragon Souvenirs, Ltd.	43	1524	172989.68
Land of Toys Inc.	49	1631	164069.44
The Sharp Gifts Warehouse	40	1656	160010.27
AV Stores, Co.	51	1778	157807.81
Anna's Decorations, Ltd	46	1469	153996.13

KPI 5: Negara dengan Penjualan Tertinggi

KPI ini mengukur performa penjualan berdasarkan geografi. USA adalah pasar terbesar dengan \$3,6 juta (~36% dari total revenue global), jauh melampaui negara lainnya. Eropa secara kolektif (Spain, France, UK, Italy, Finland, Norway, Denmark) menyumbang porsi besar berikutnya, menunjukkan bahwa EMEA adalah wilayah strategis kedua. Insight ini berguna untuk perencanaan ekspansi pasar dan alokasi tenaga penjualan per wilayah.

Query :

SELECT

l.country AS negara,

l.territory AS wilayah,

COUNT(f.id_factOrder) AS jumlah_transaksi,

SUM(f.sales) AS total_revenue

FROM fact_orders f

JOIN dimlocation l ON f.id_dimLocation = l.id_dimLocation

GROUP BY l.country, l.territory

ORDER BY total_revenue DESC;

Hasil :

Showing rows 0 - 18 (19 total, Query took 0.0102 seconds.)

```
SELECT l.country AS negara, l.territory AS wilayah, COUNT(f.id_factOrder) AS jumlah_transaksi, SUM(f.sales) AS total_revenue FROM fact_orders f JOIN dimlocation l ON f.id_dimLocation = l.id_dimLocation GROUP BY l.country, l.territory ORDER BY total_revenue DESC;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

negara	wilayah	jumlah_transaksi	total_revenue
USA	NA	1004	3627982.83
Spain	EMEA	342	1215686.92
France	EMEA	314	1110916.52
Australia	APAC	185	630623.10
UK	EMEA	144	478880.46
Italy	EMEA	113	374674.31
Finland	EMEA	92	329581.91
Norway	EMEA	85	307463.70
Singapore	Japan	79	288488.41
Denmark	EMEA	63	245637.15
Canada	NA	70	224078.56
Germany	EMEA	62	220472.09
Sweden	EMEA	57	210014.21
Austria	EMEA	55	202062.53
Japan	Japan	52	188167.81
Switzerland	EMEA	31	117713.56
Belgium	EMEA	33	108412.62
Philippines	Japan	26	94015.73
Ireland	EMEA	16	57756.43

V. KESIMPULAN

Berdasarkan hasil implementasi proyek ETL dan pembangunan Data Warehouse GlobalScale Retail, dapat disimpulkan bahwa seluruh tahapan proyek mulai dari pengambilan data, perancangan skema, hingga implementasi pipeline ETL telah berhasil dilaksanakan dengan baik.

Data Warehouse yang dibangun menggunakan model Star Schema dengan satu tabel fakta dan empat tabel dimensi berhasil menampung dan mengorganisasi 2.823 baris data transaksi dari sumber CSV menjadi struktur yang terorganisir dan siap dianalisis. Proses ETL yang dirancang mampu menangani pembersihan data, konversi format tanggal, deduplikasi, serta pembentukan surrogate key untuk menjaga integritas relasi antar tabel.

Dari hasil analisis lima KPI yang dilakukan terhadap Data Warehouse, diperoleh kesimpulan bahwa perusahaan mengalami pertumbuhan revenue yang positif pada periode 2003–2004, dengan Classic Cars sebagai lini produk paling dominan, Medium Deal sebagai segmen transaksi terbesar, serta Amerika Serikat sebagai pasar utama yang menyumbang lebih dari sepertiga total pendapatan global. Selain itu, teridentifikasi dua pelanggan utama yang memberikan kontribusi signifikan terhadap revenue perusahaan sehingga perlu mendapat perhatian khusus dalam strategi retensi pelanggan.

Dengan demikian, Data Warehouse ini terbukti dapat menjadi fondasi yang kuat untuk mendukung proses analisis data dan pengambilan keputusan bisnis berbasis data secara efisien dan terstruktur.

Link github : https://github.com/nideeuw/Project-UAS_Kelompok-4.git